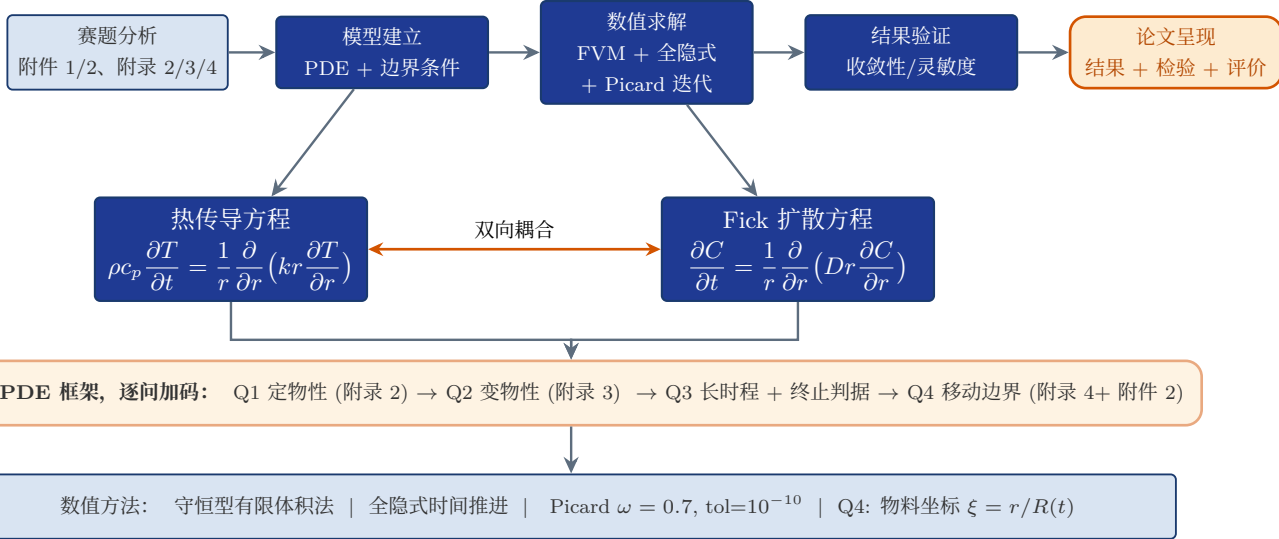
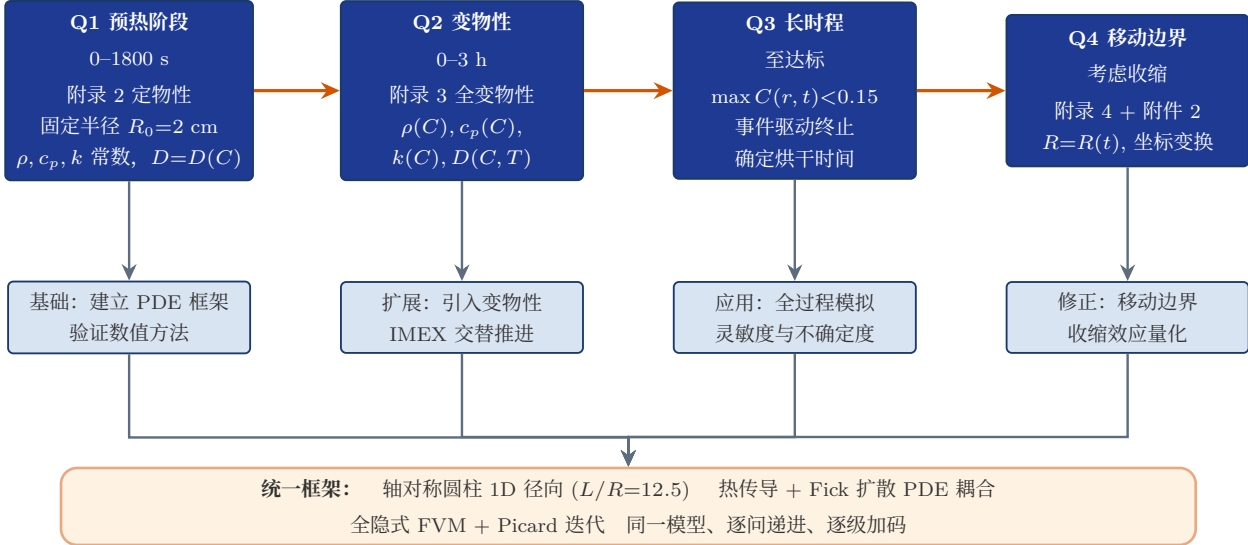
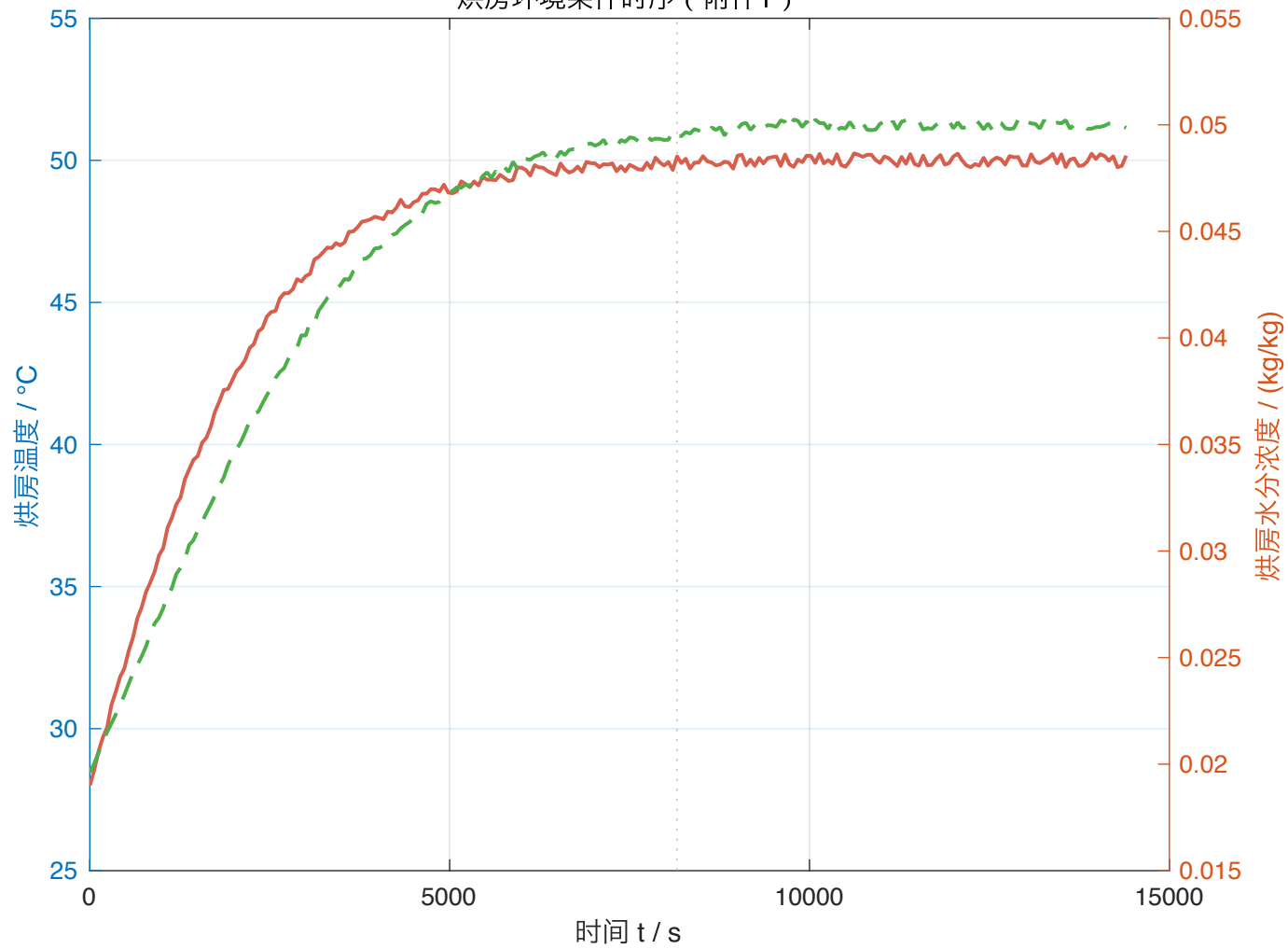


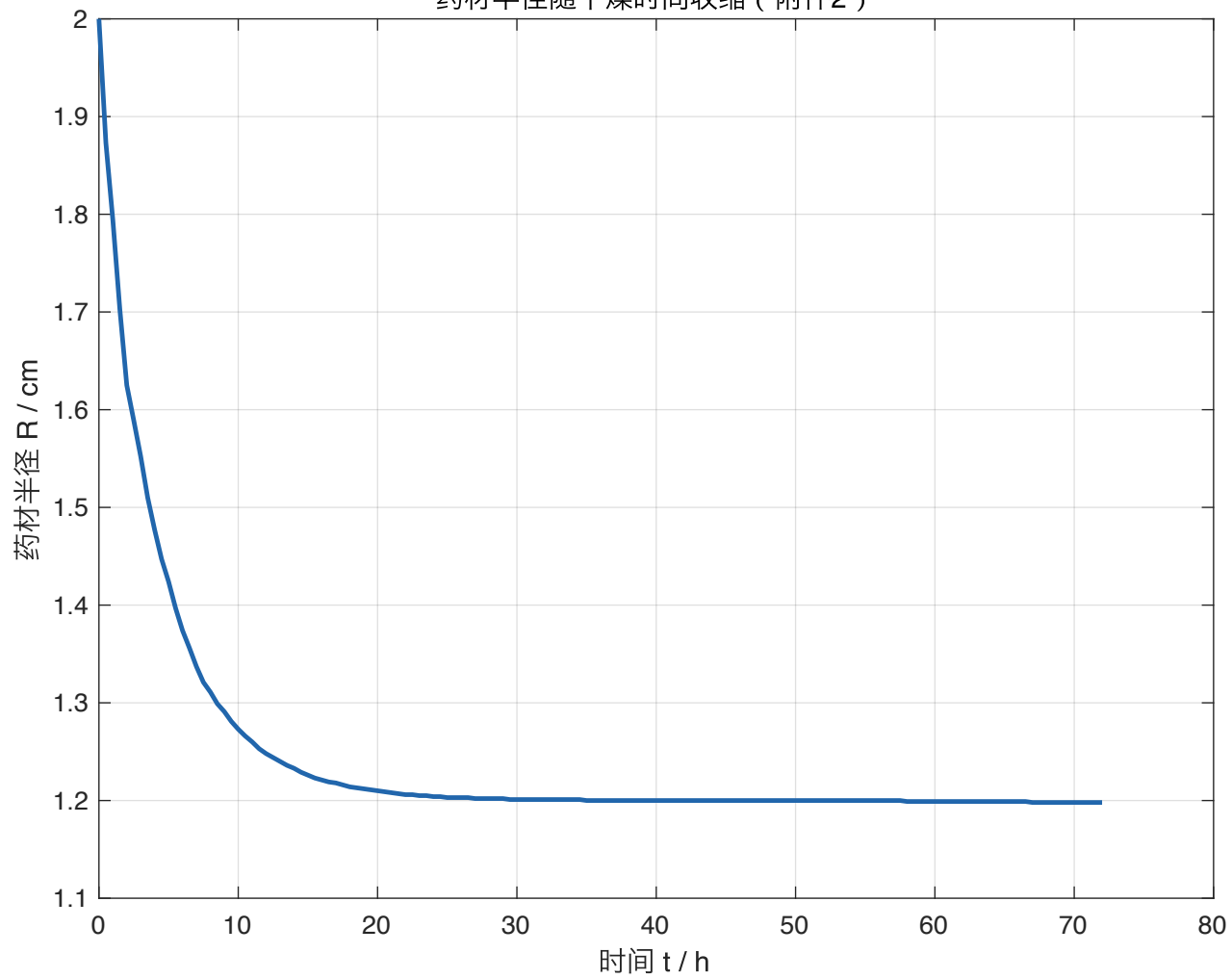




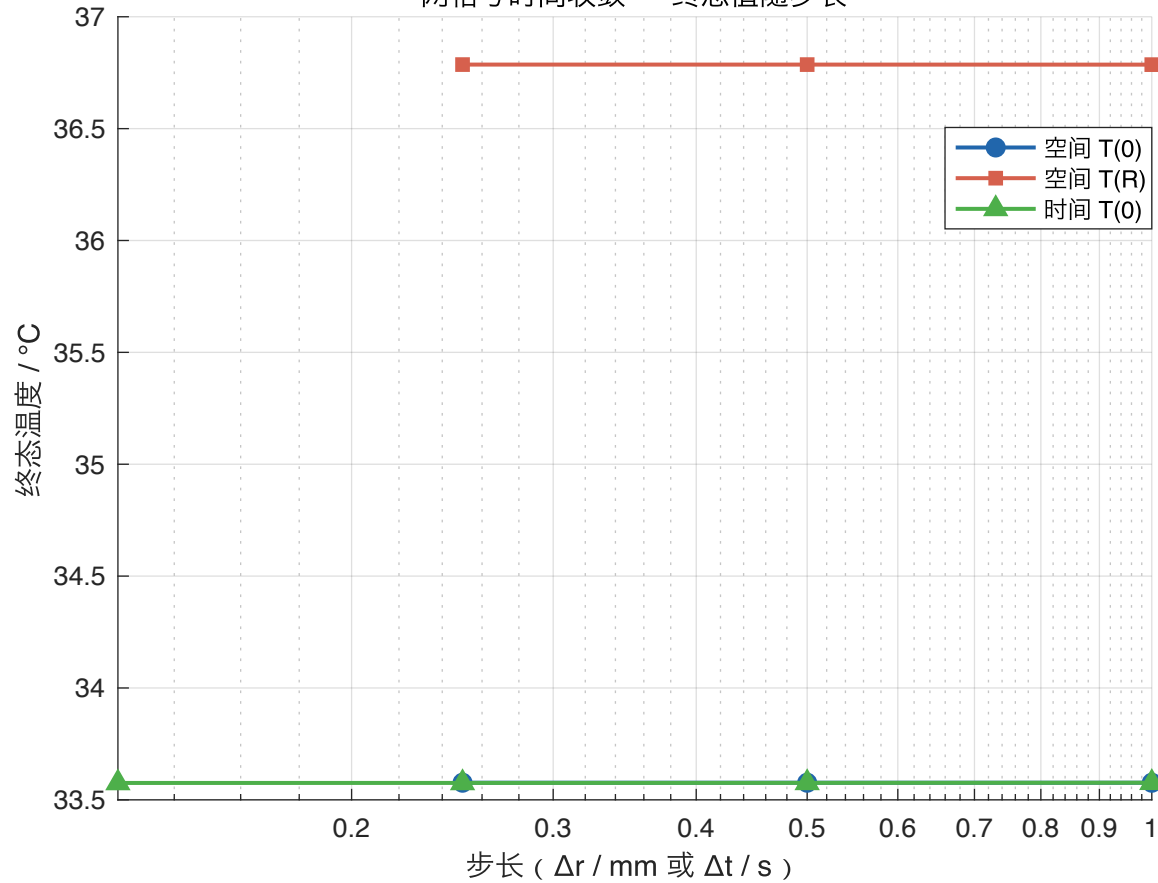
烘房环境条件时序 (附件1)



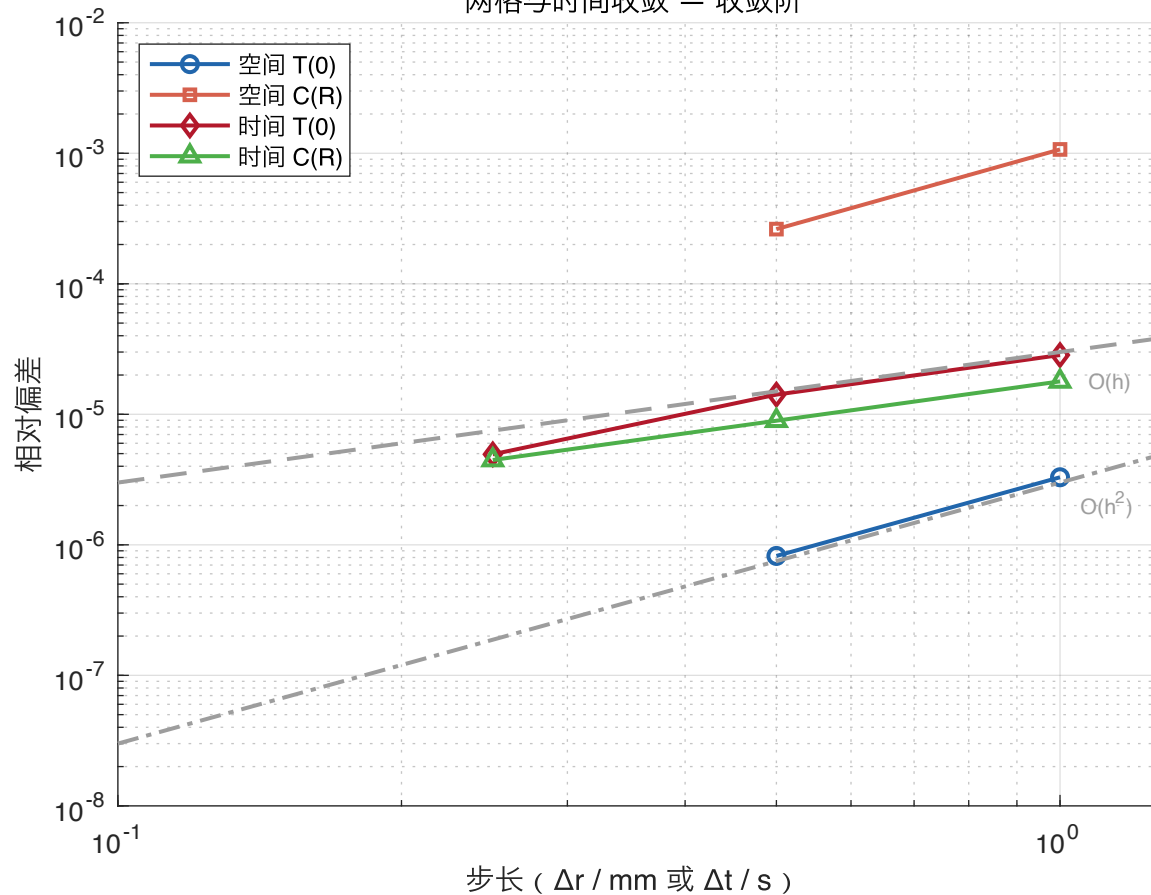
药材半径随干燥时间收缩 (附件2)

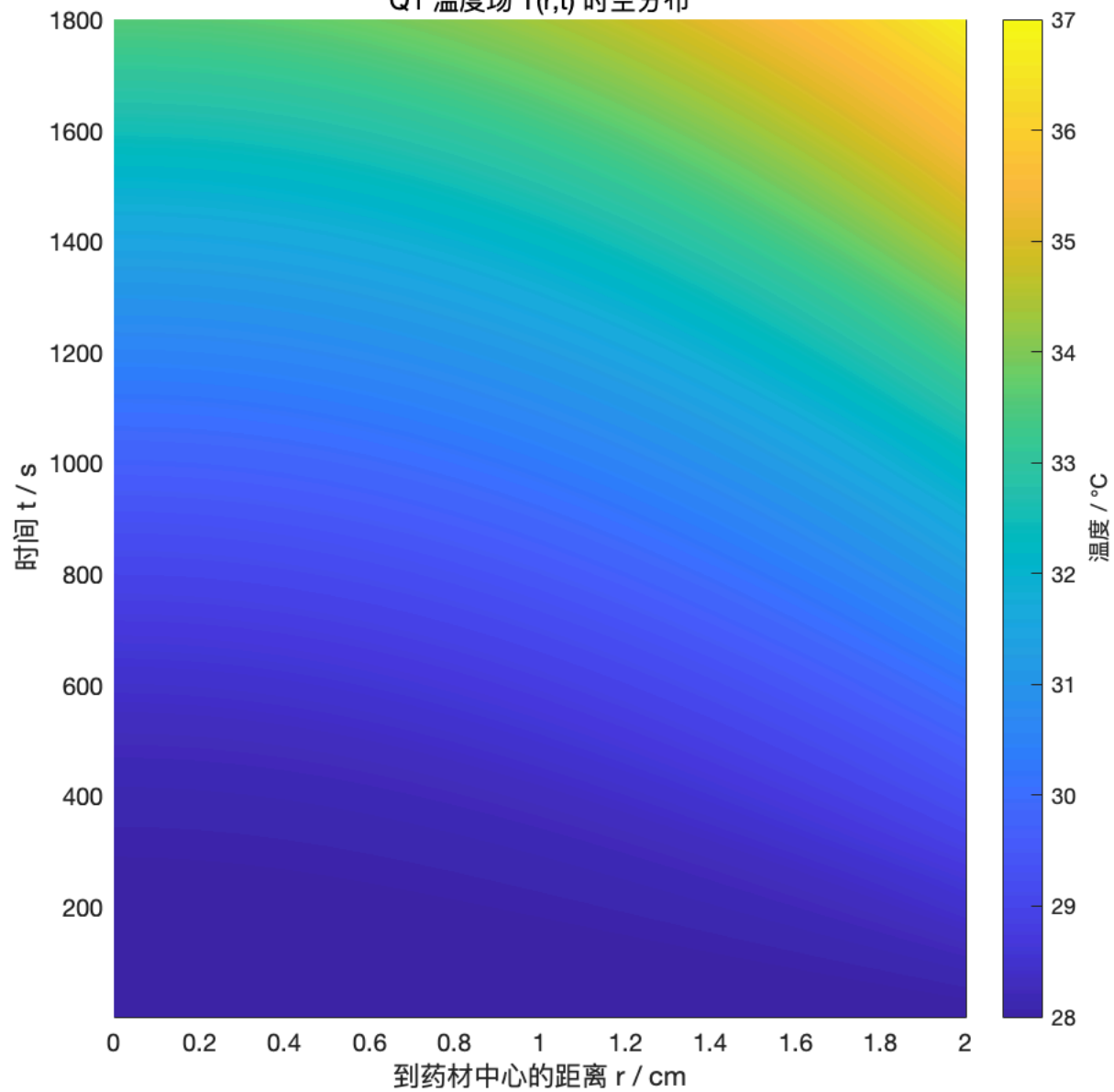


网格与时间收敛 — 终态值随步长

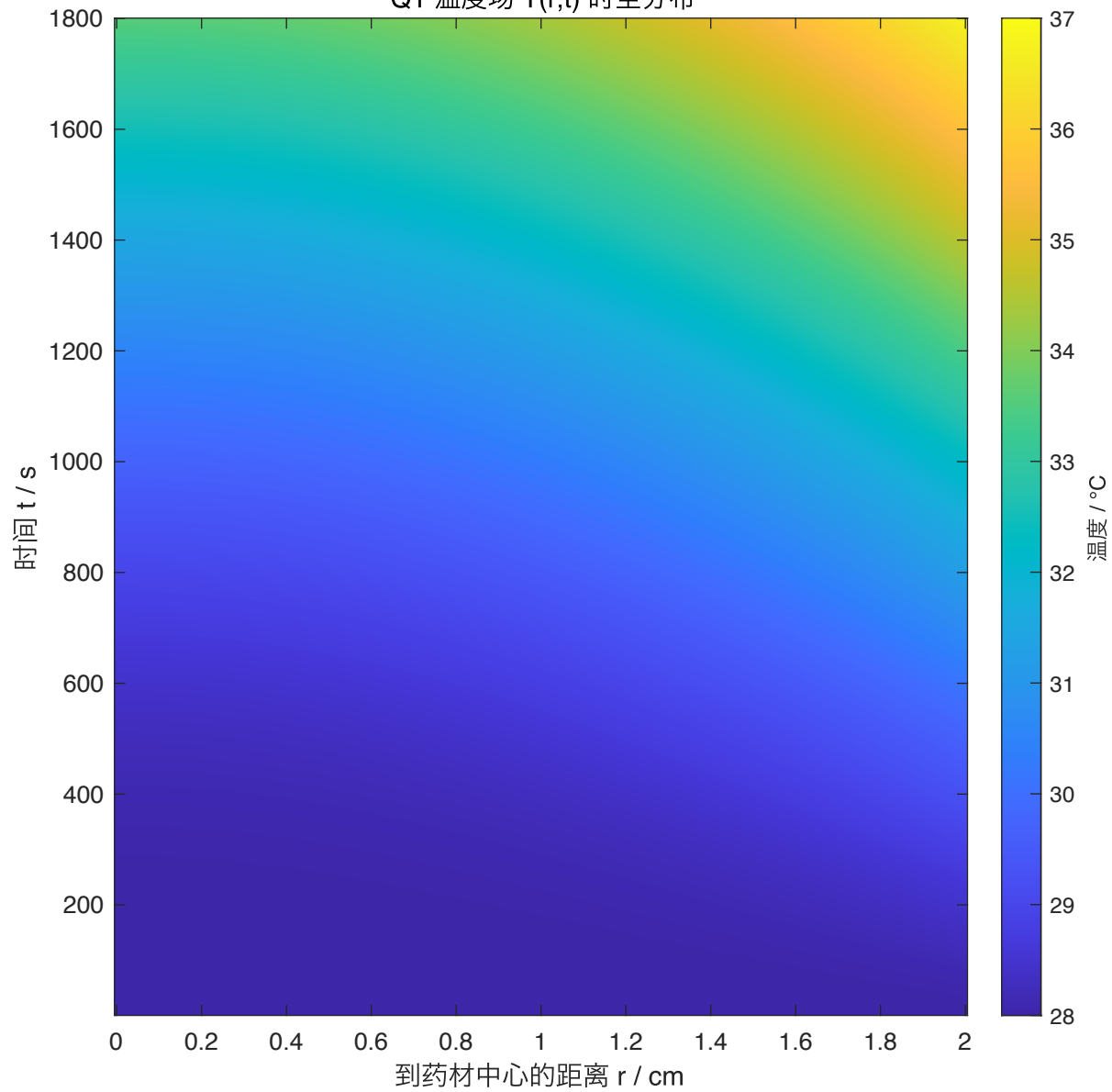


网格与时间收敛 — 收敛阶

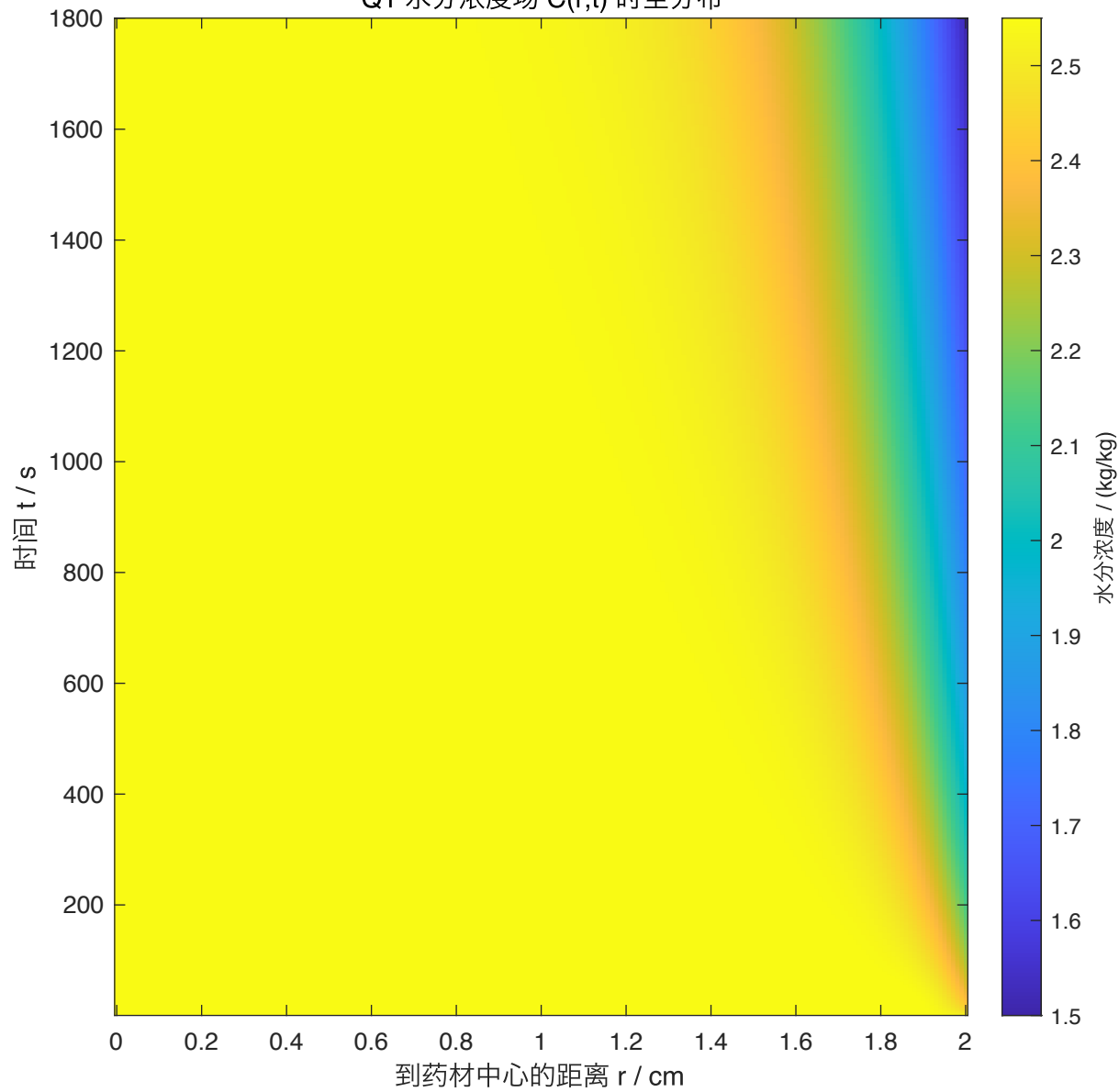


Q1 温度场 $T(r,t)$ 时空分布

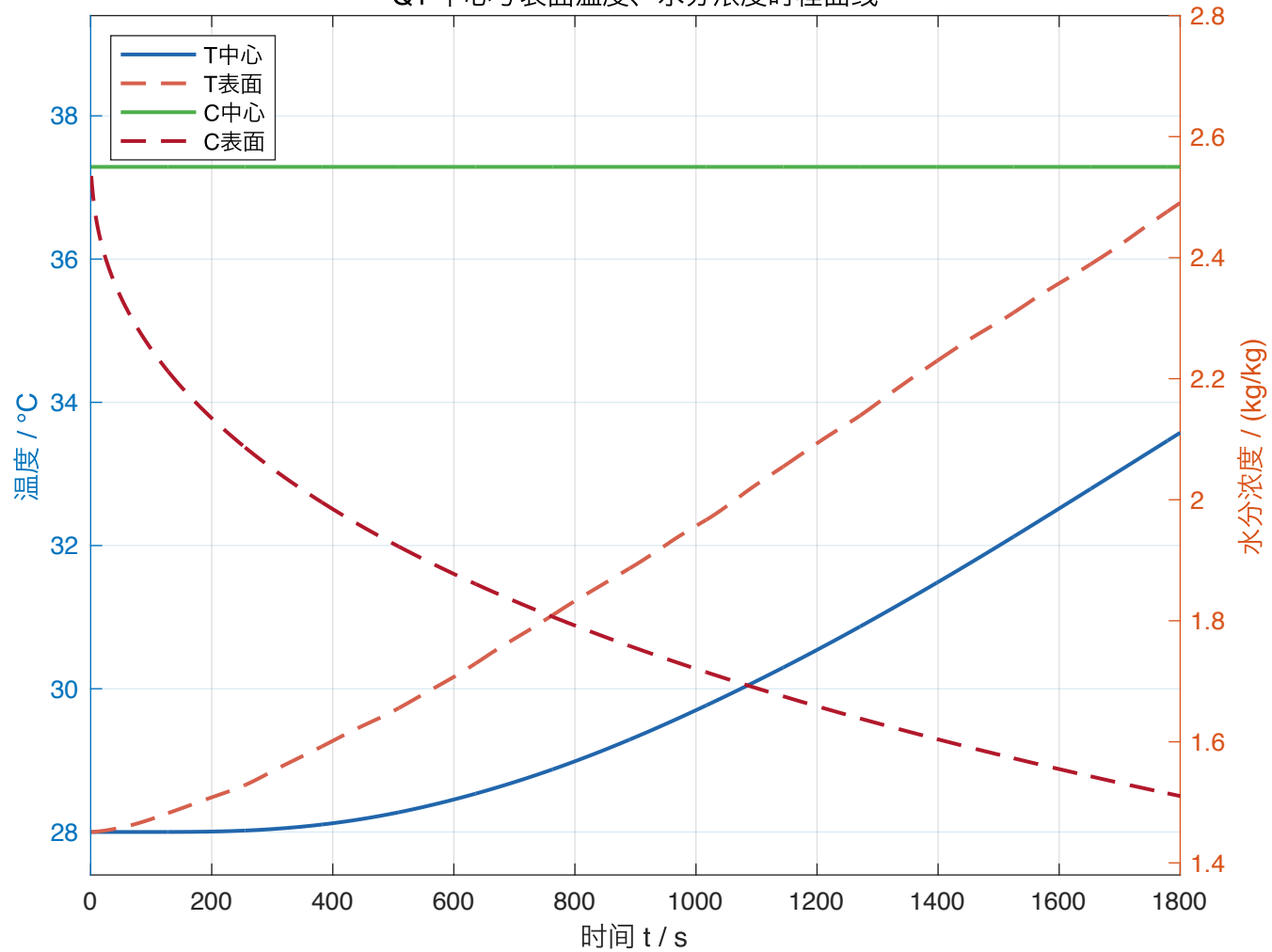
Q1 温度场 $T(r,t)$ 时空分布



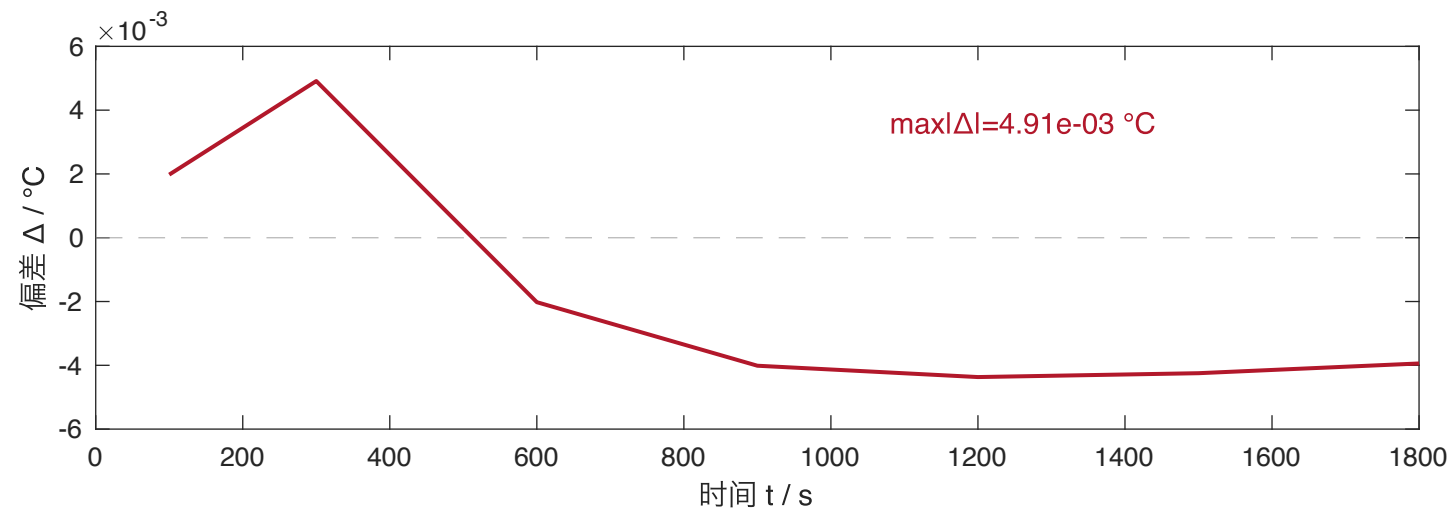
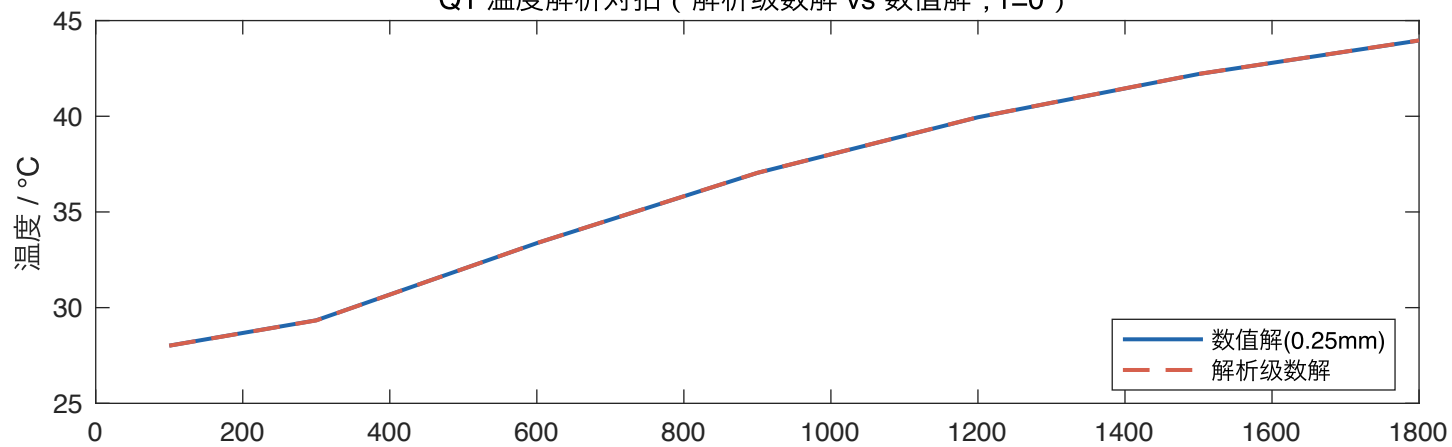
Q1 水分浓度场 $C(r,t)$ 时空分布

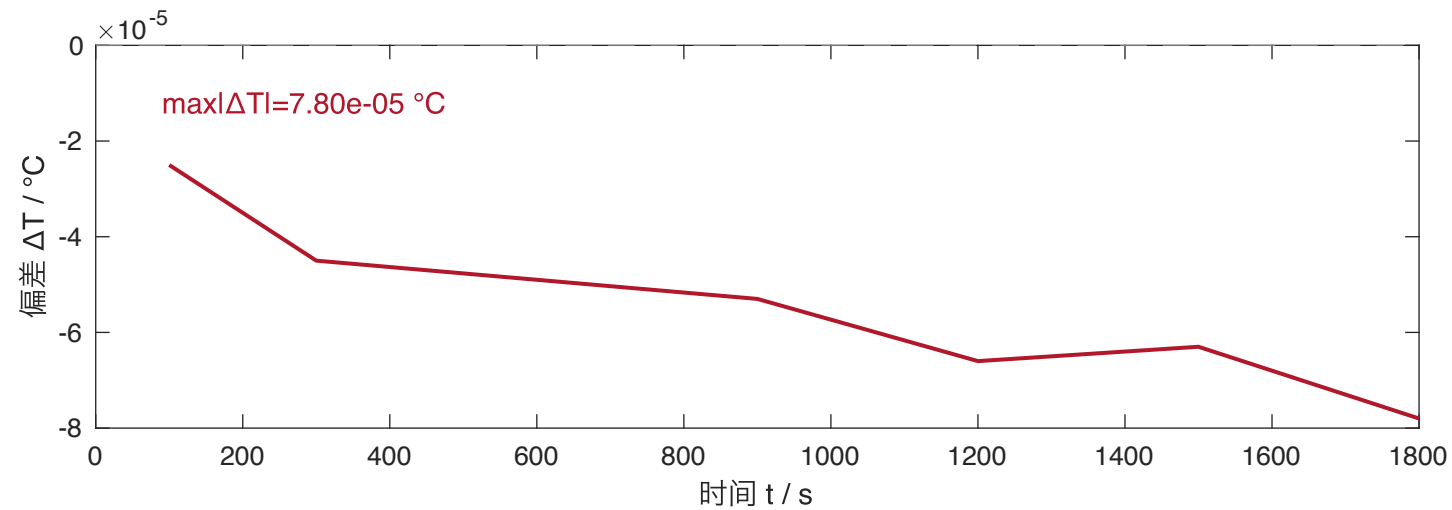
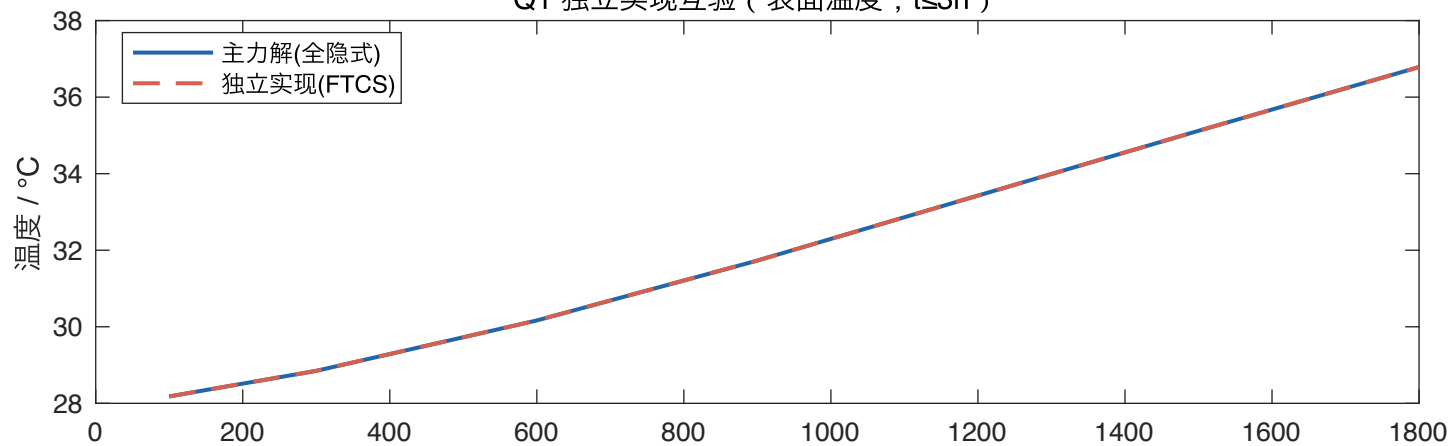


Q1 中心与表面温度、水分浓度时程曲线



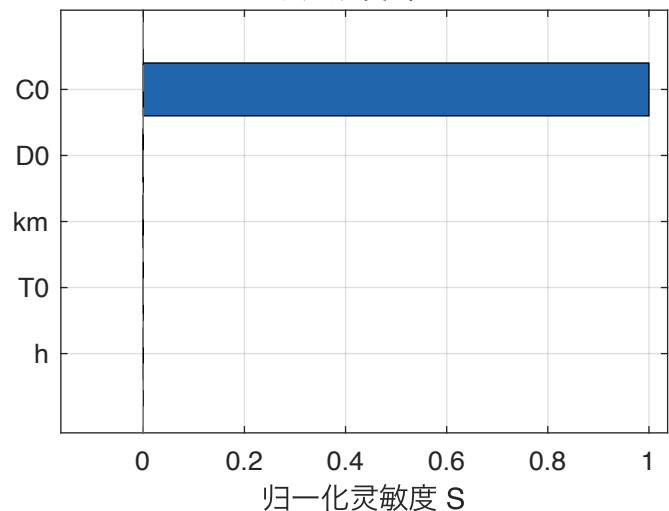
Q1 温度解析对拍 (解析级数解 vs 数值解, $r=0$)



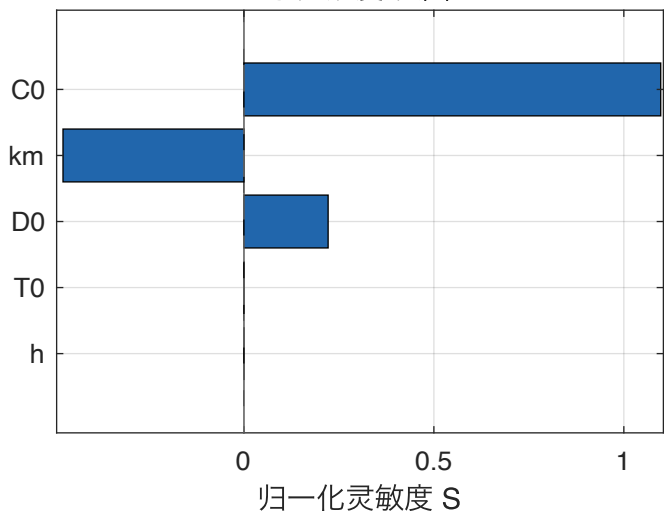
Q1 独立实现互验 (表面温度, $t \leq 3h$)

Q1 参数灵敏度龙卷风图 (OAT±20%)

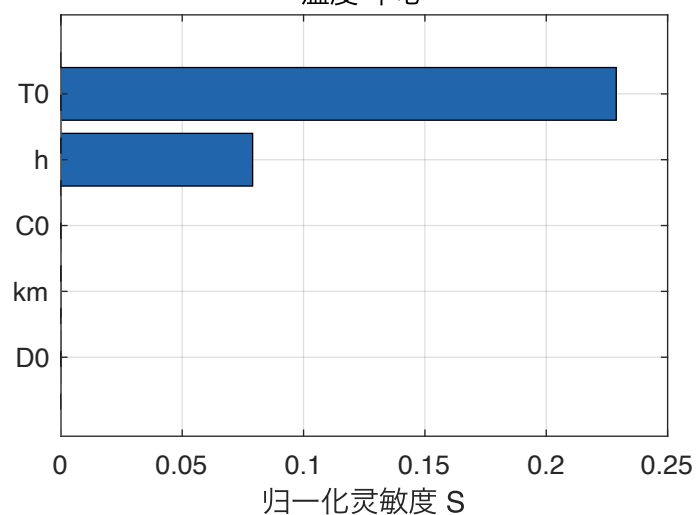
水分浓度 中心



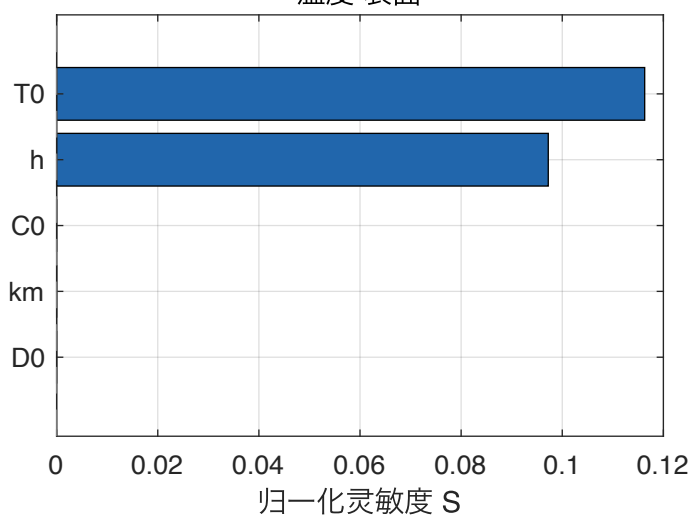
水分浓度 表面



温度 中心

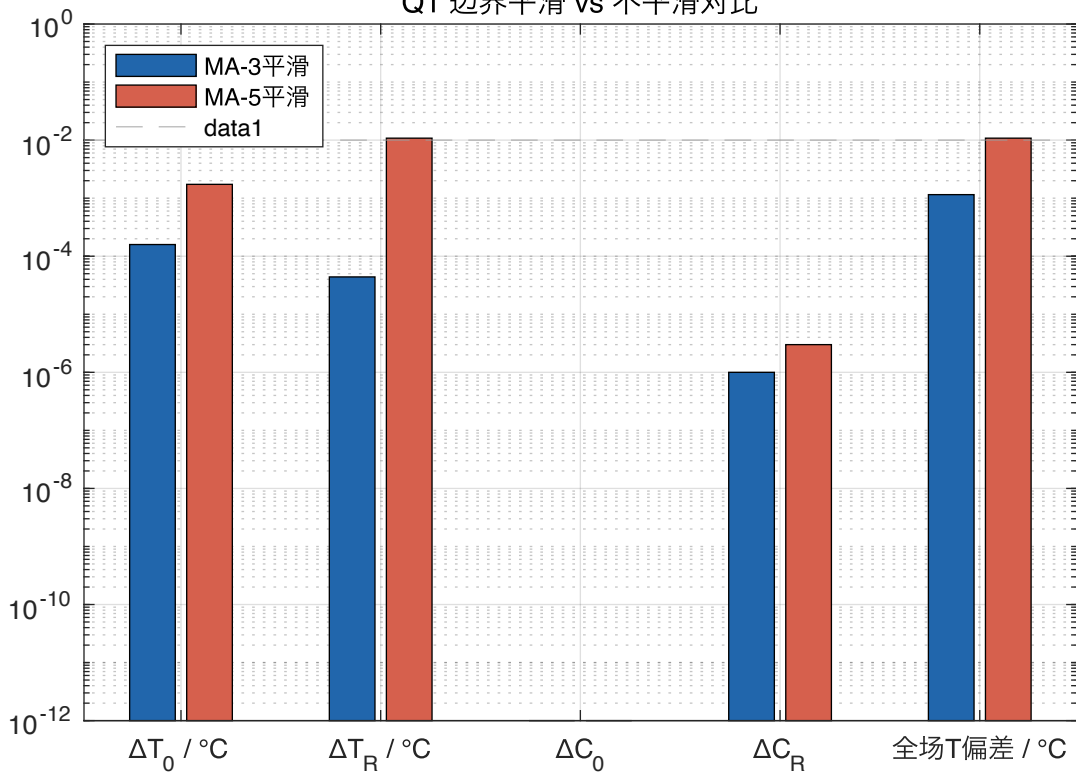


温度 表面

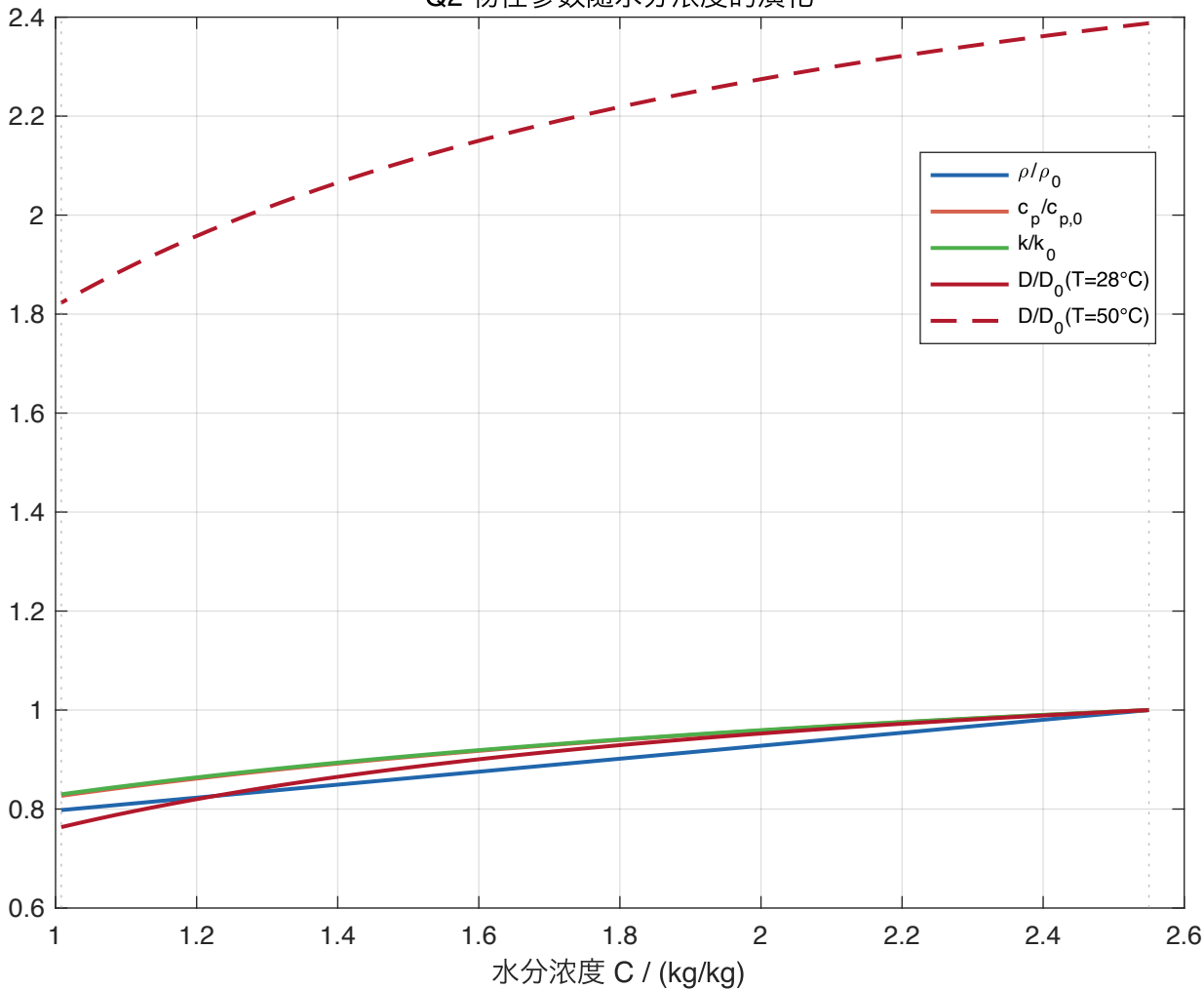


Q1 边界平滑 vs 不平滑对比

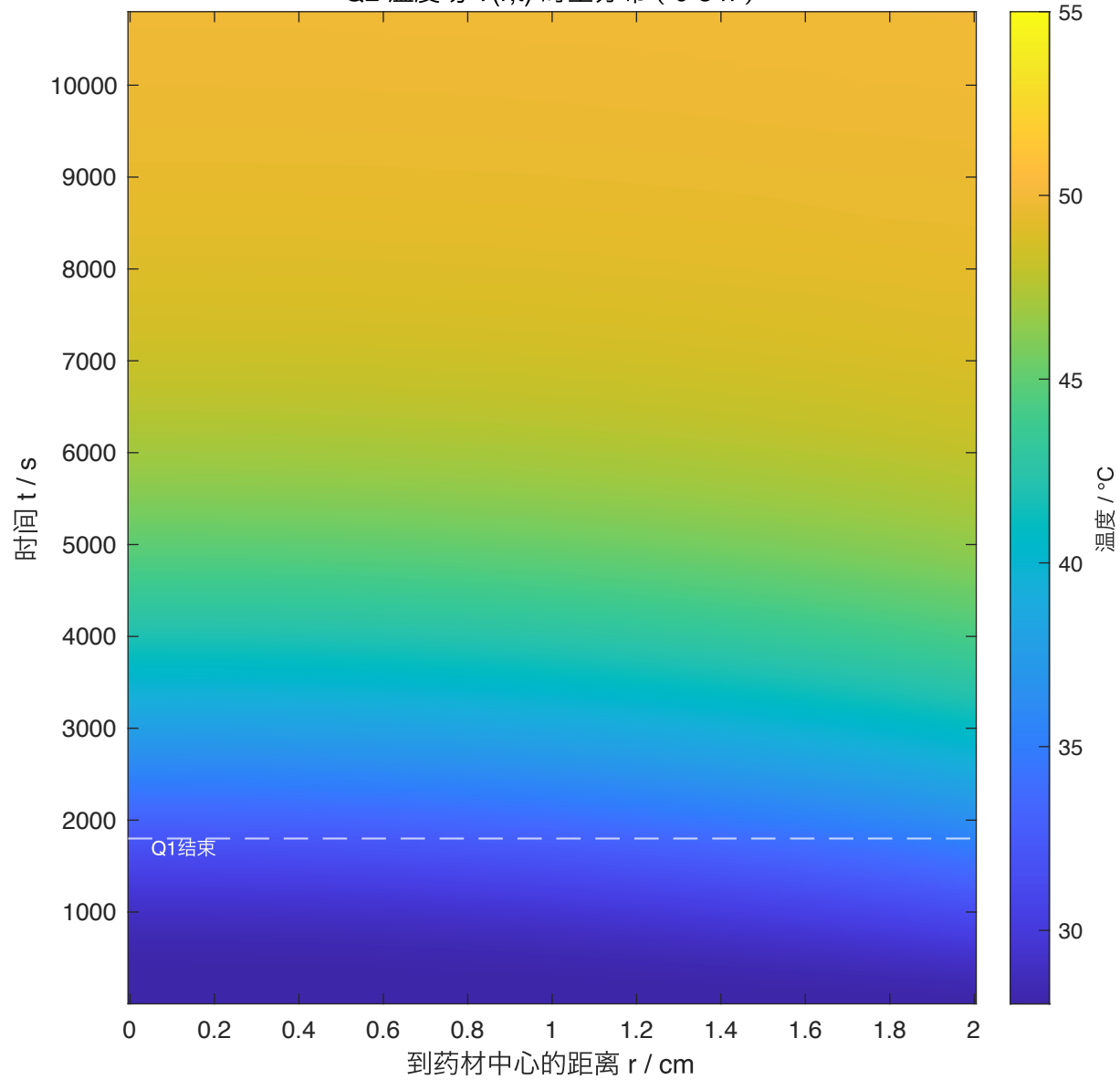
偏差绝对值 (对数轴)



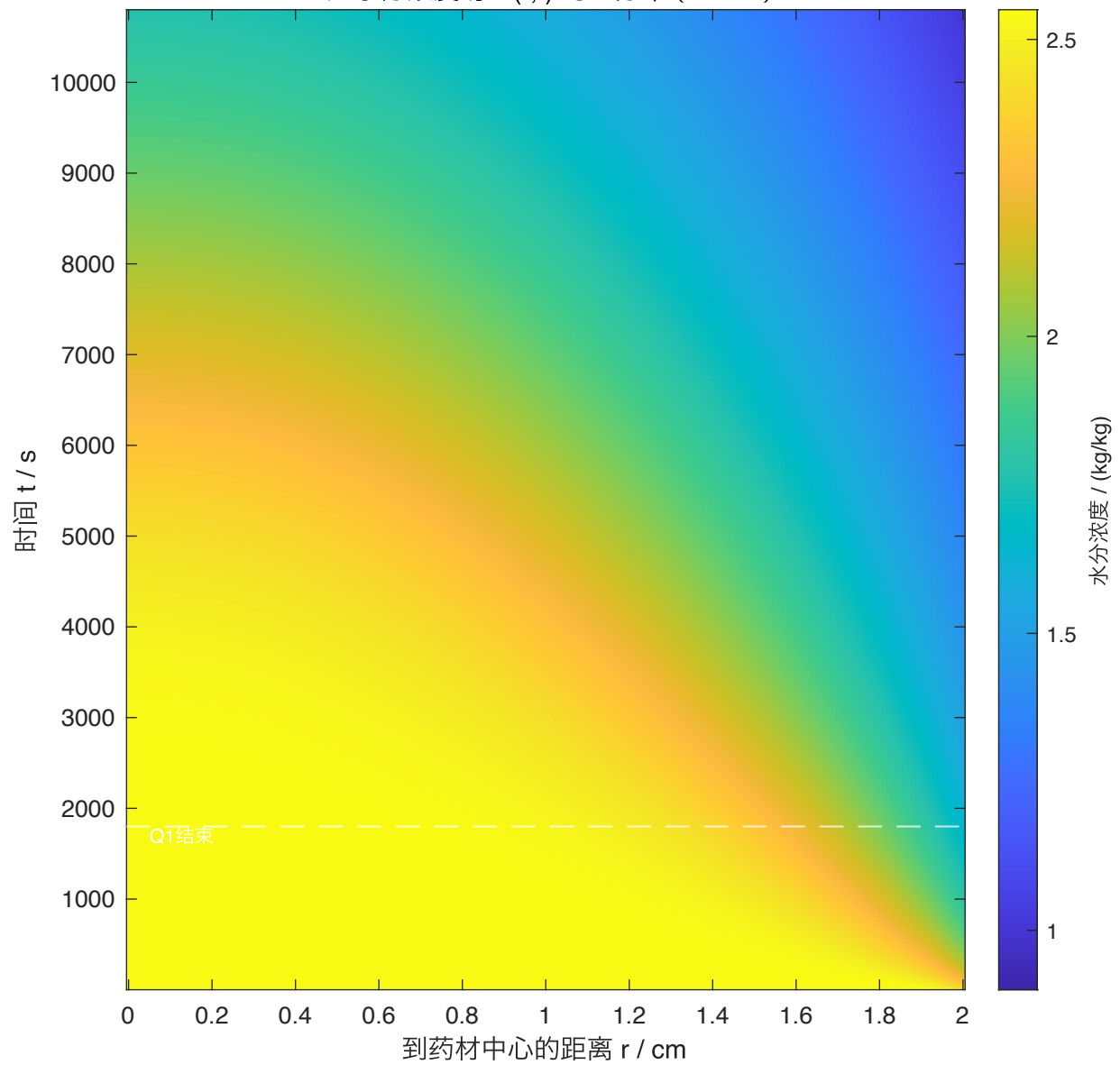
Q2 物性参数随水分浓度的演化

相对初值 ($C_0=2.55$ 处=1)水分浓度 C / (kg/kg)

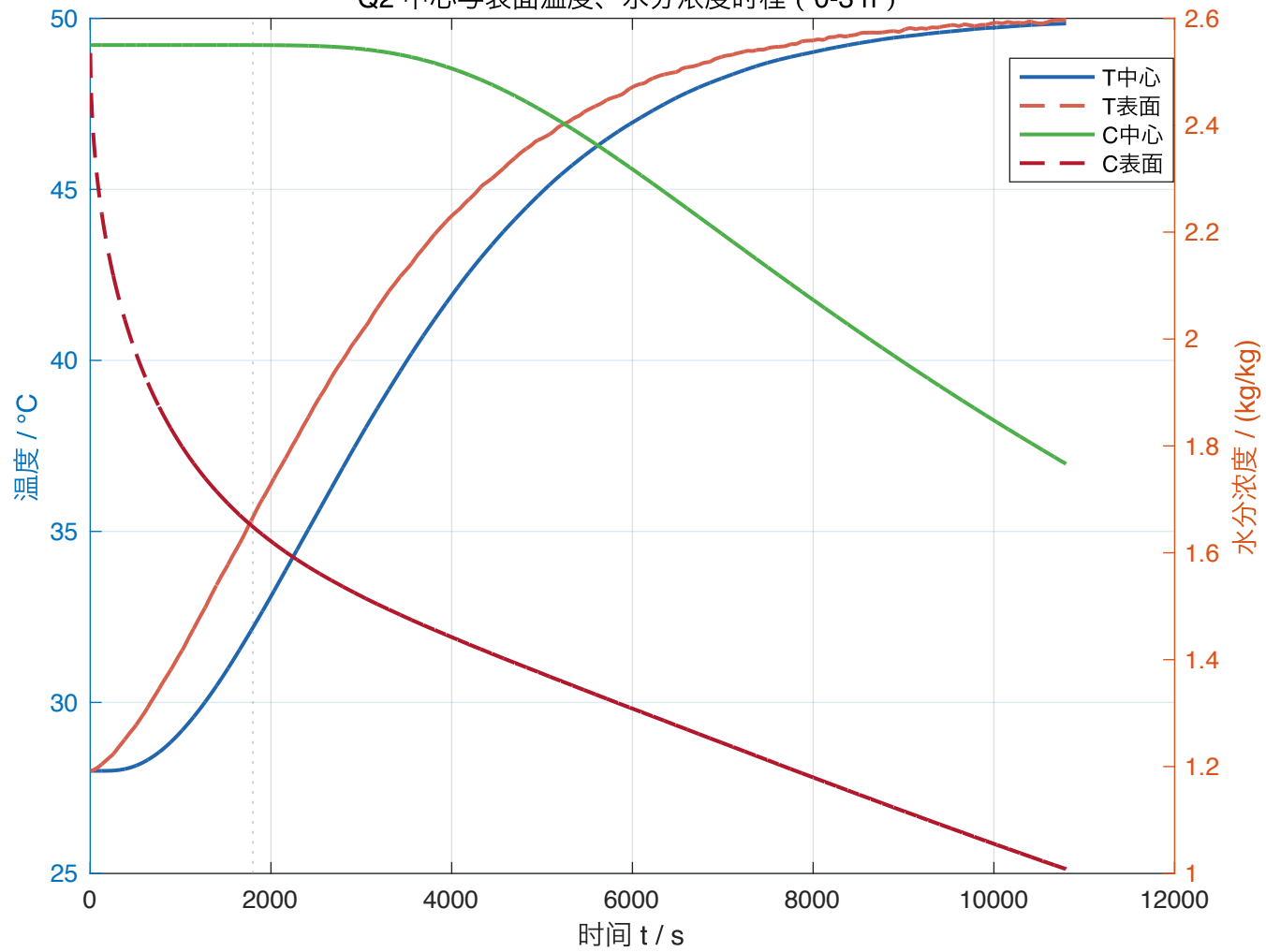
Q2 温度场 $T(r,t)$ 时空分布 (0-3 h)



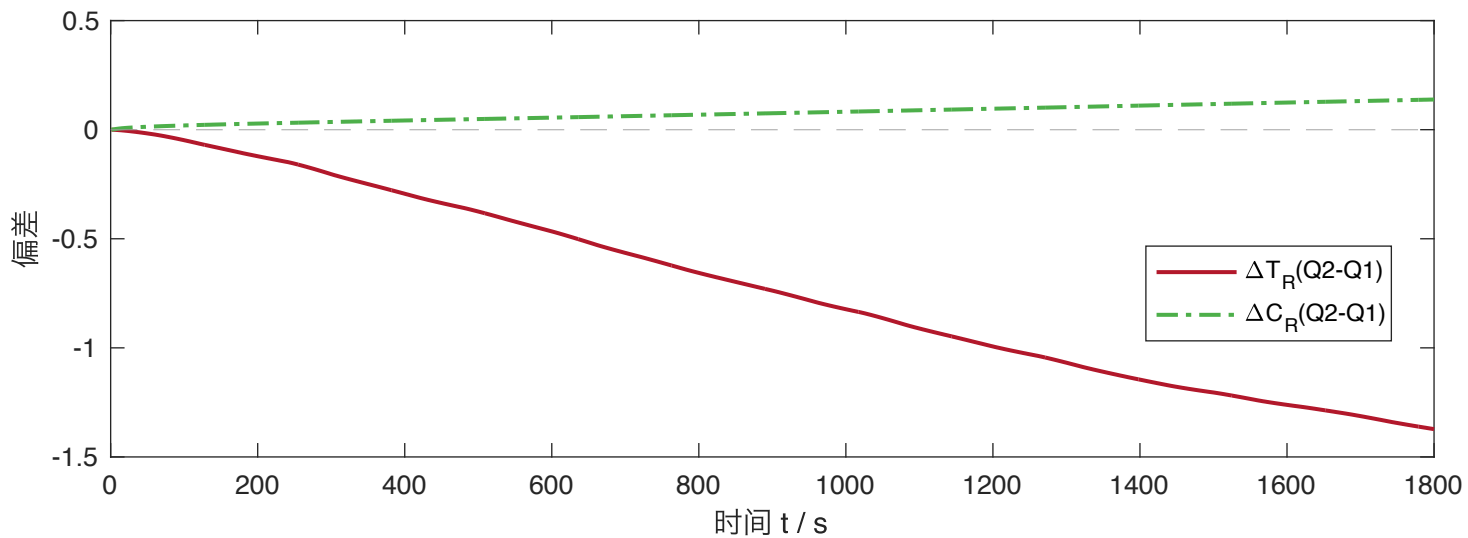
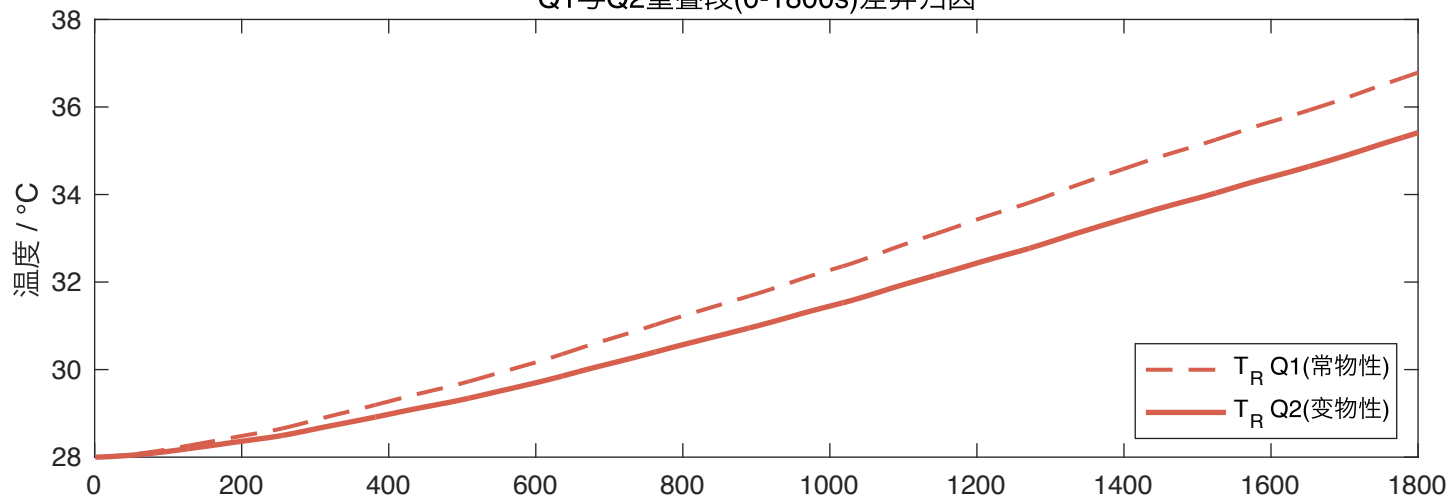
Q2 水分浓度场 $C(r,t)$ 时空分布 (0-3 h)



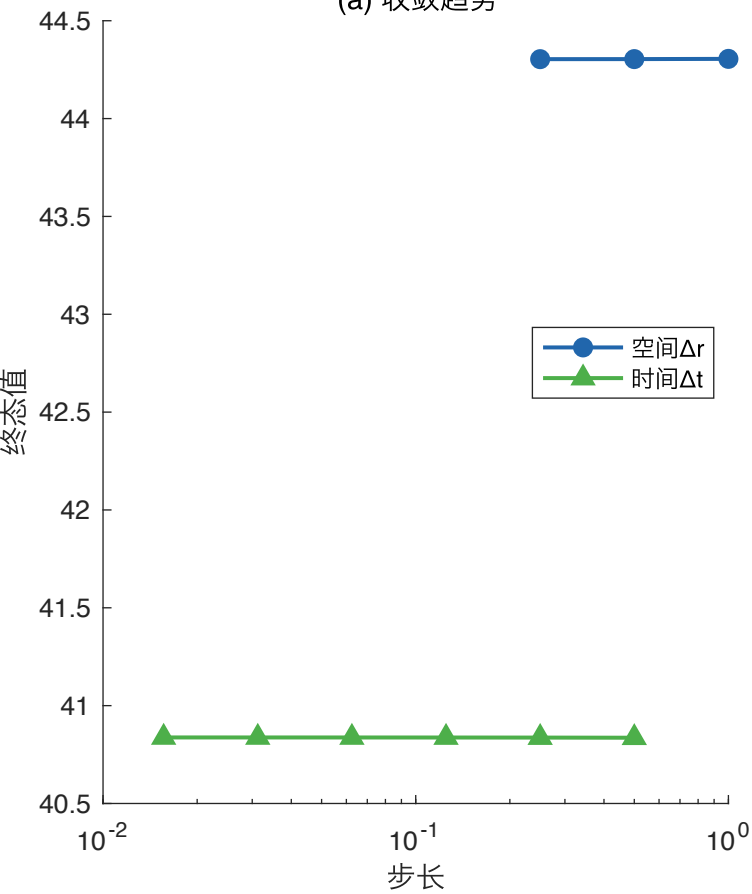
Q2 中心与表面温度、水分浓度时程 (0-3 h)



Q1与Q2重叠段(0-1800s)差异归因



(a) 收敛趋势



(b) 收敛阶

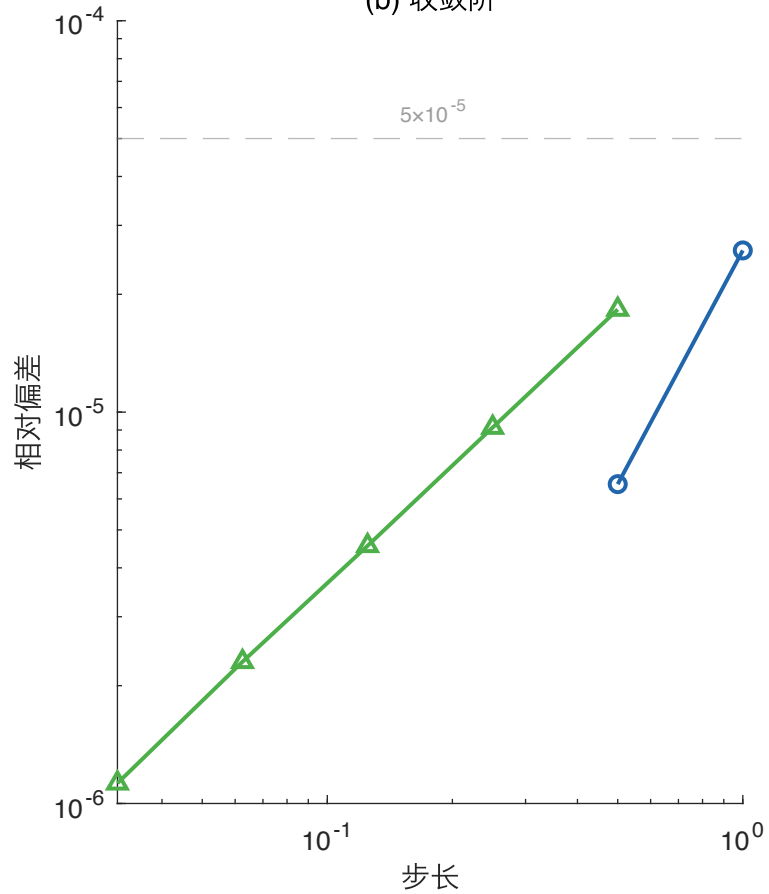
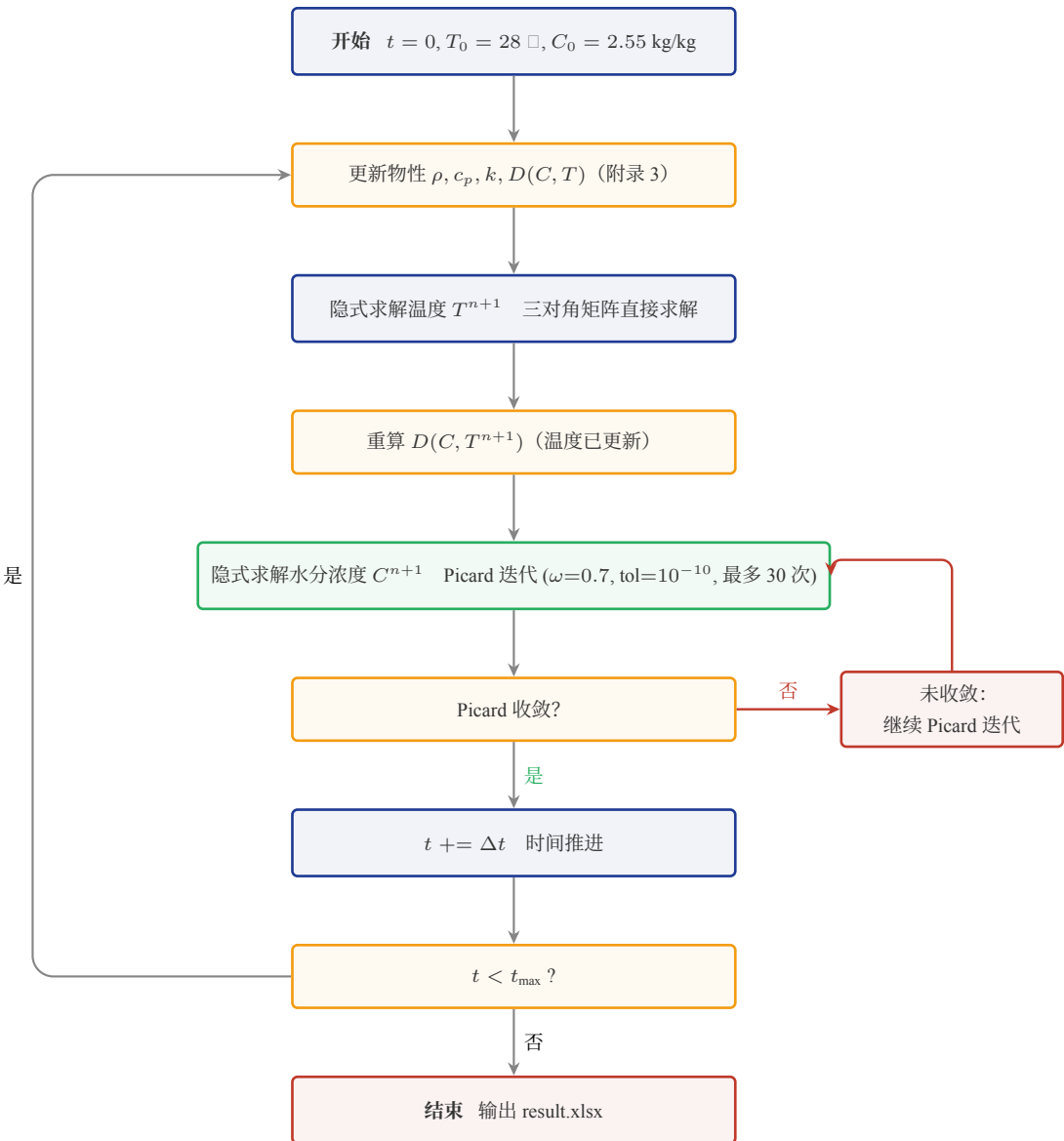
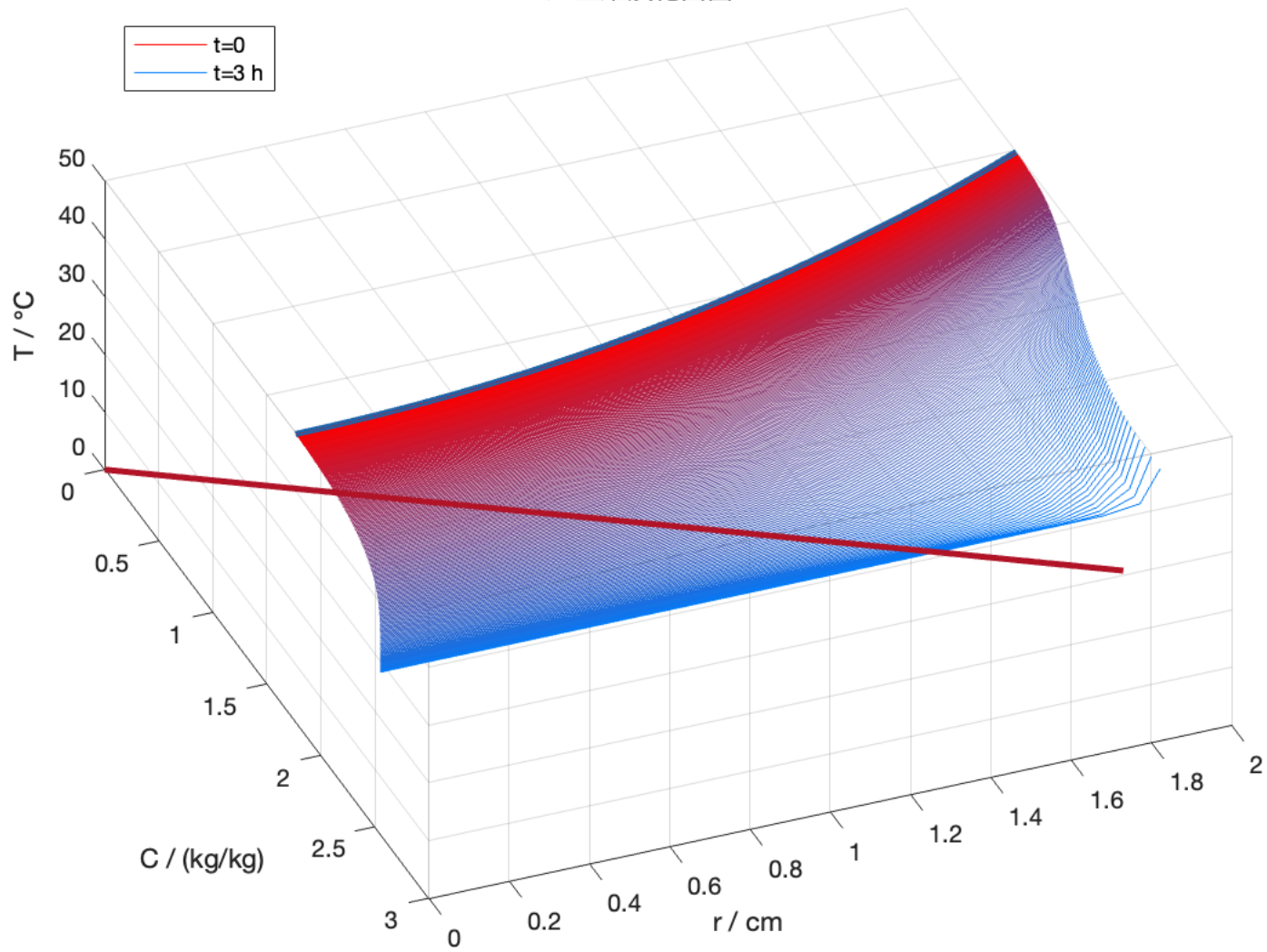


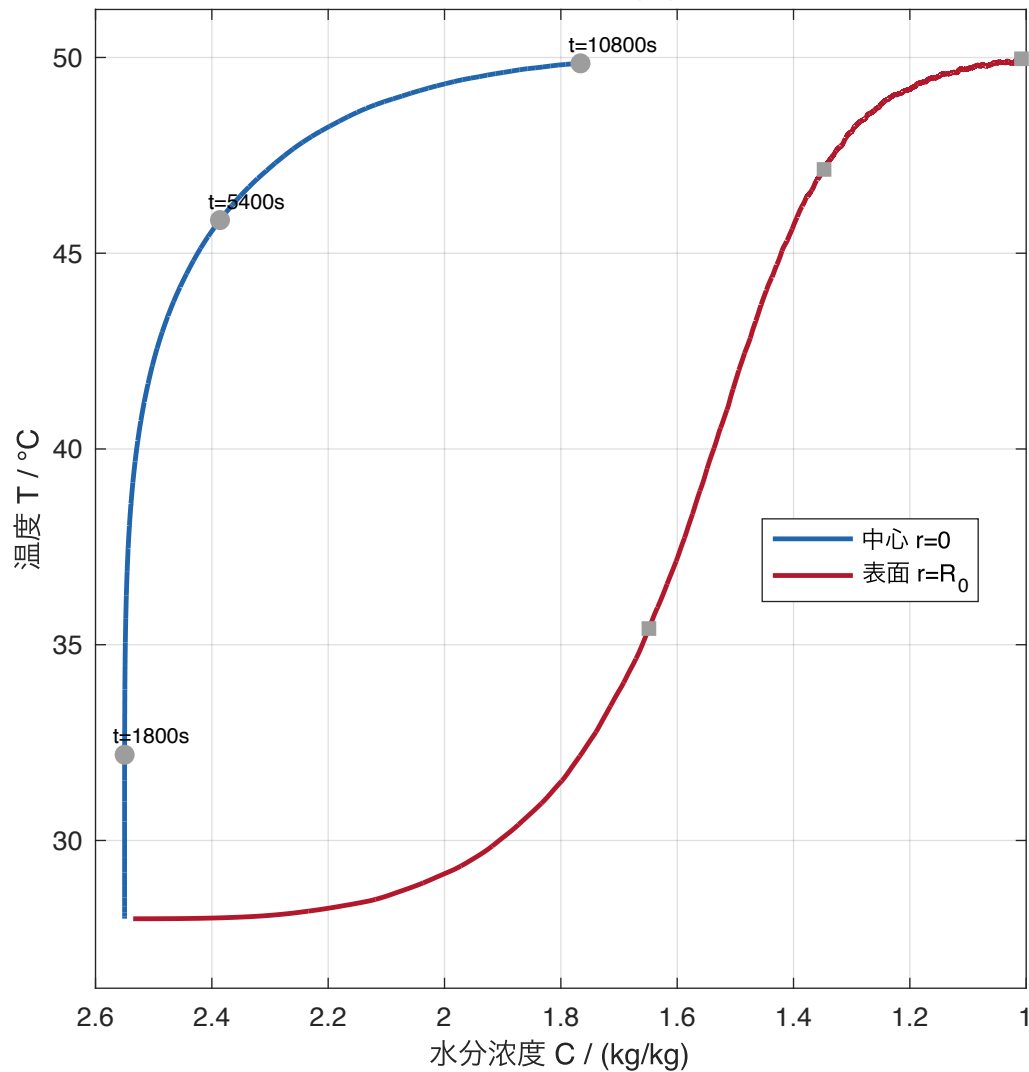
图 22: IMEX 交替推进算法流程



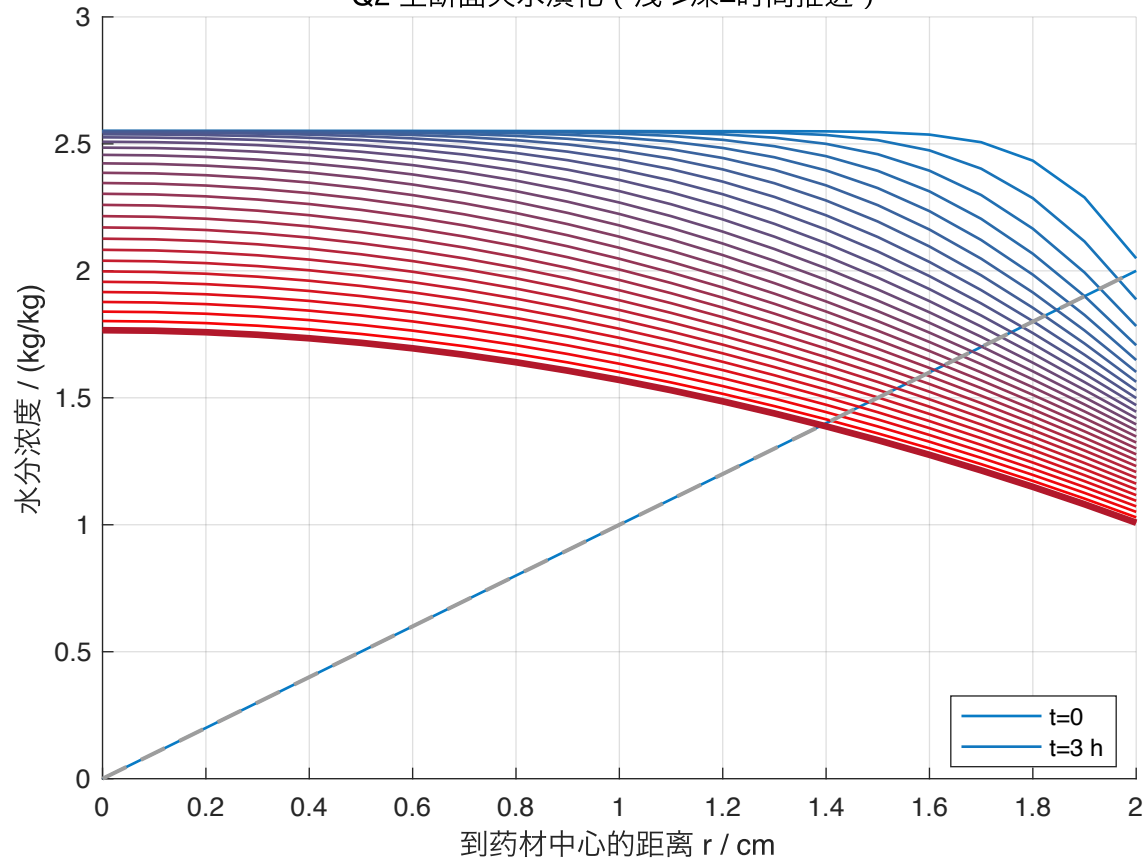
Q2 三维演化曲面 T-C-t



Q2 温度-水分浓度相轨迹

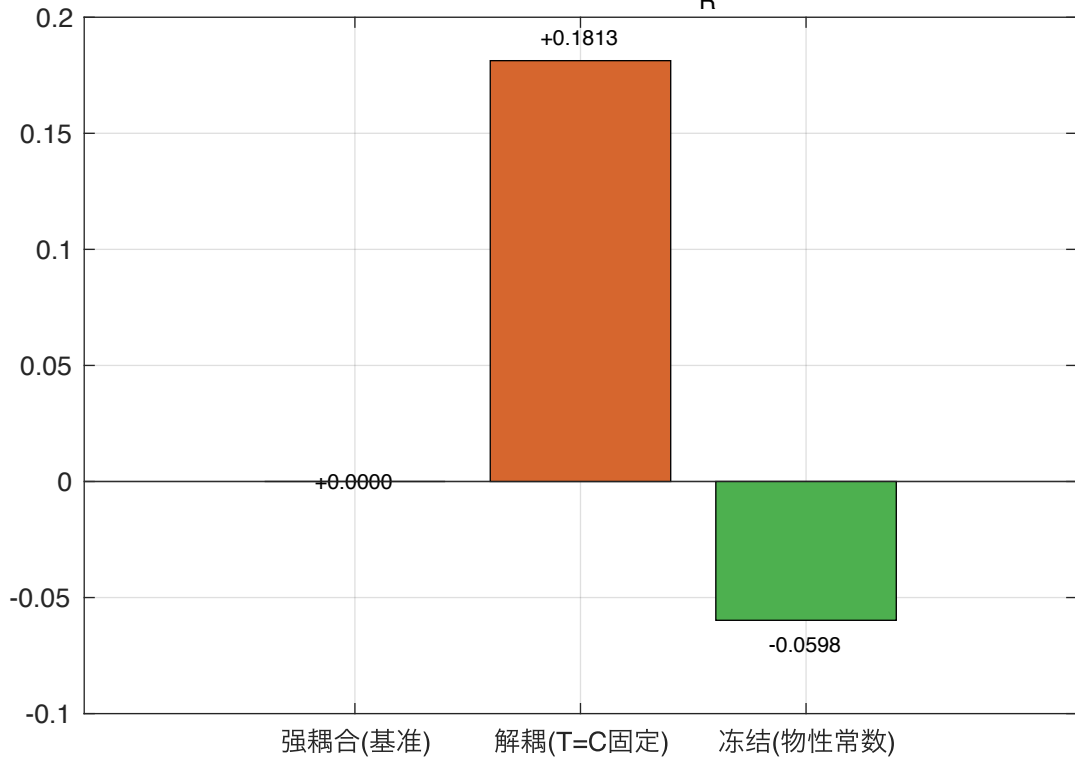


Q2 全断面失水演化 (浅->深=时间推进)

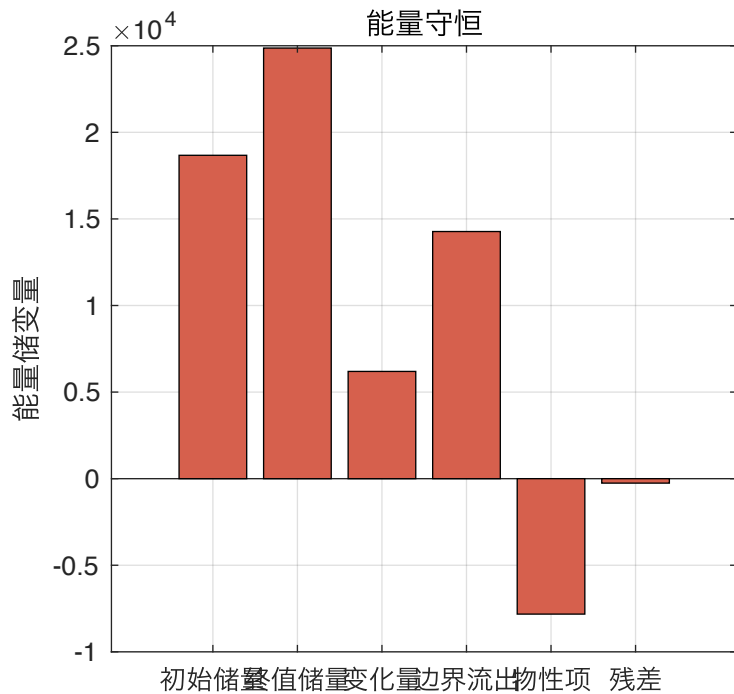
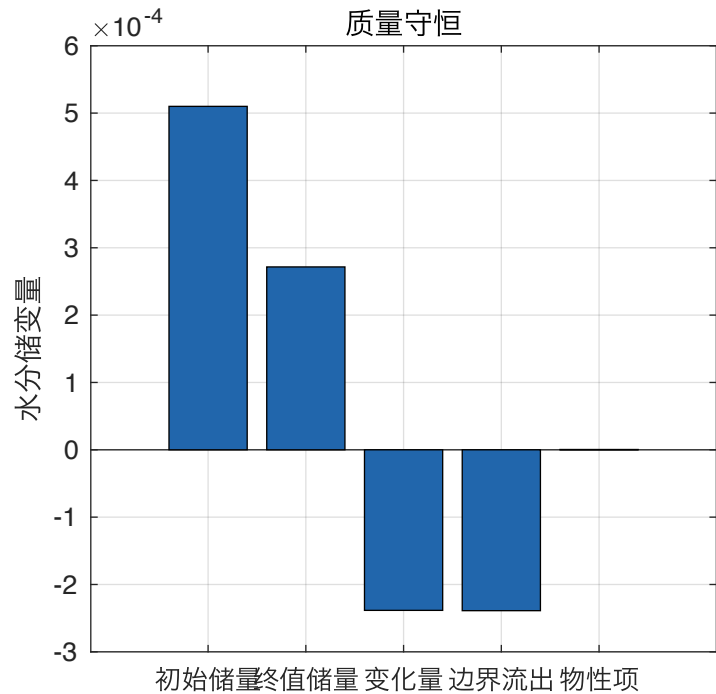


Q2 耦合强度分解 (3 h末 C_R 偏差)

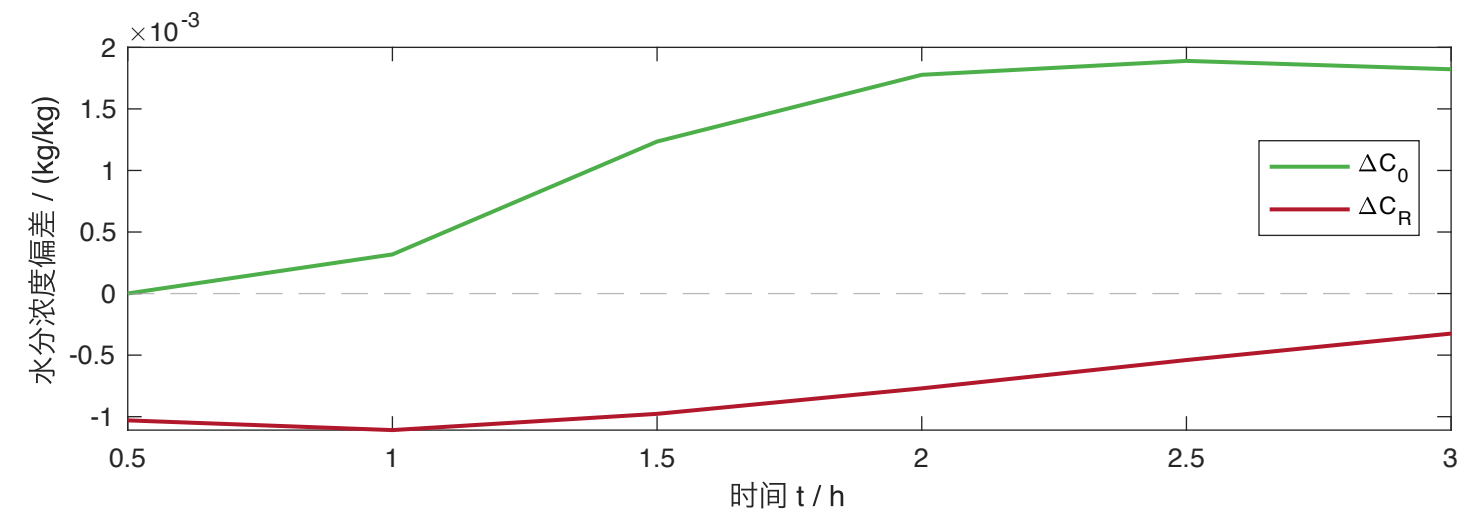
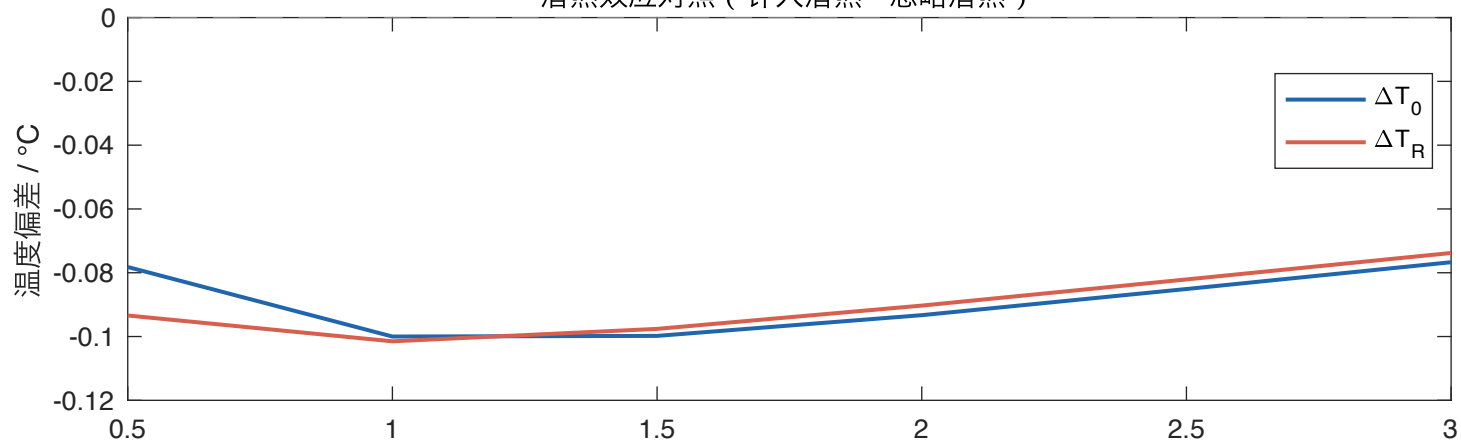
相对偏差 (以强耦合 C_R 为基准)



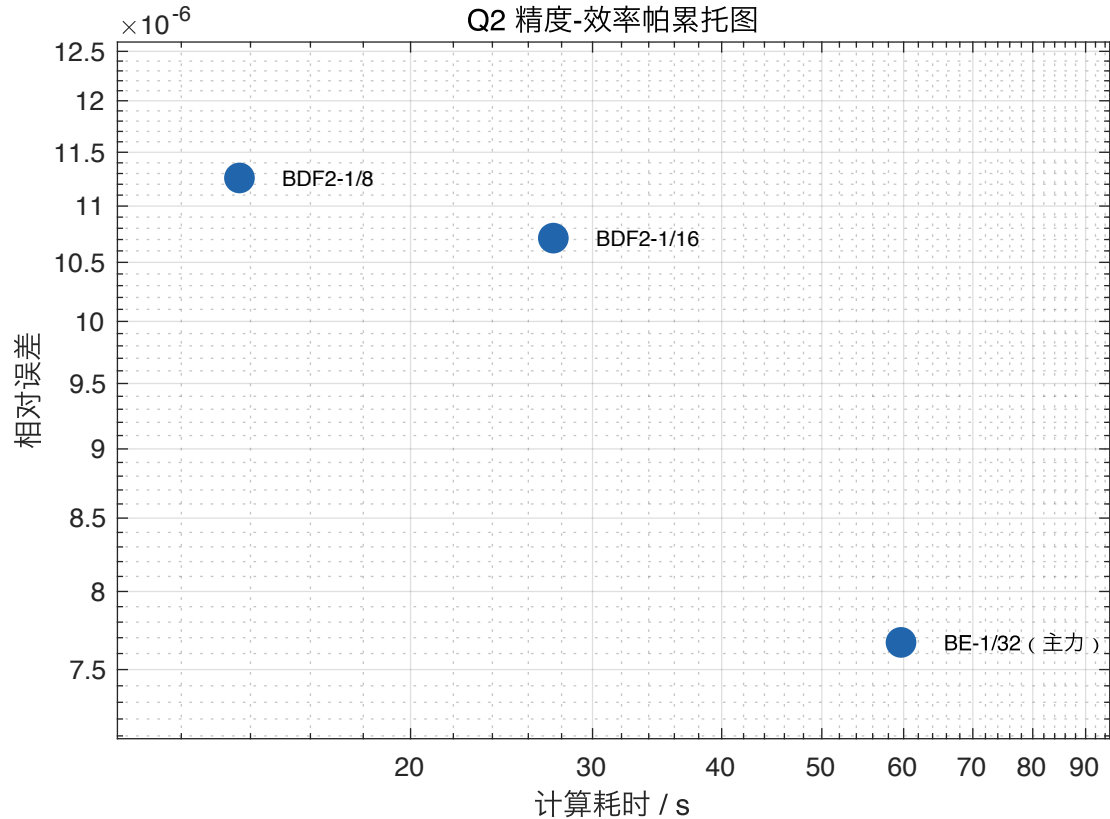
Q2 守恒性总账

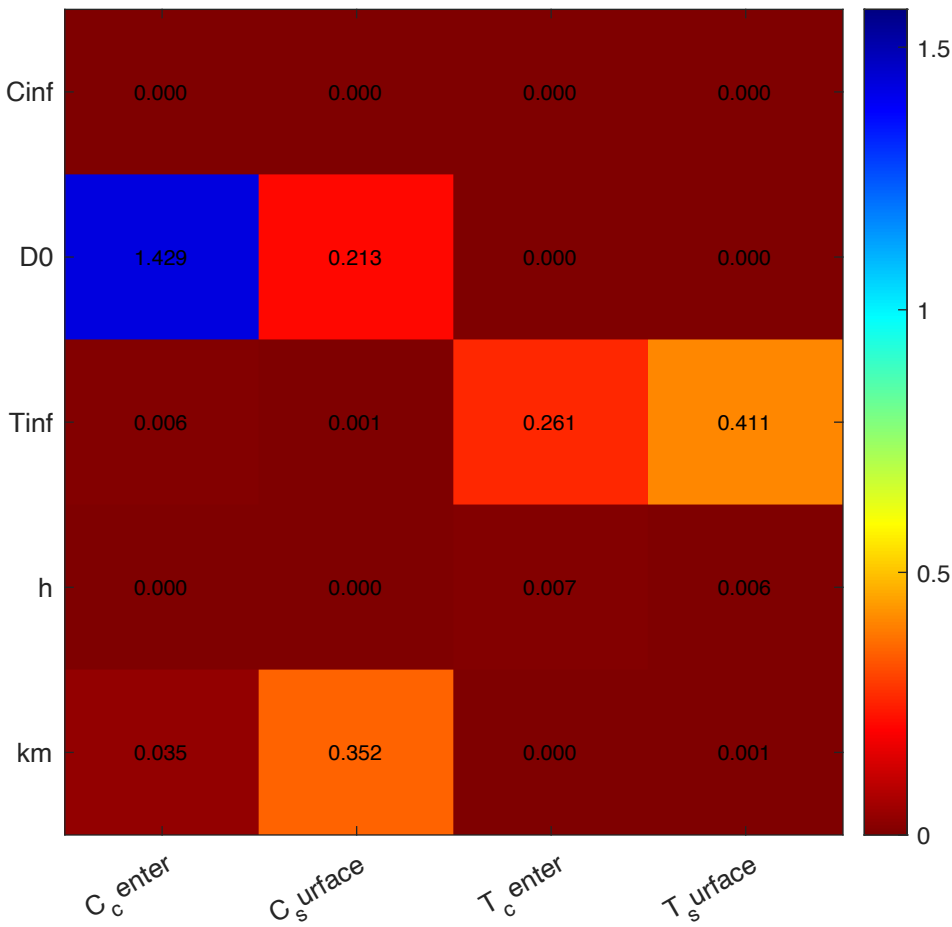


潜热效应对照 (计入潜热 - 忽略潜热)

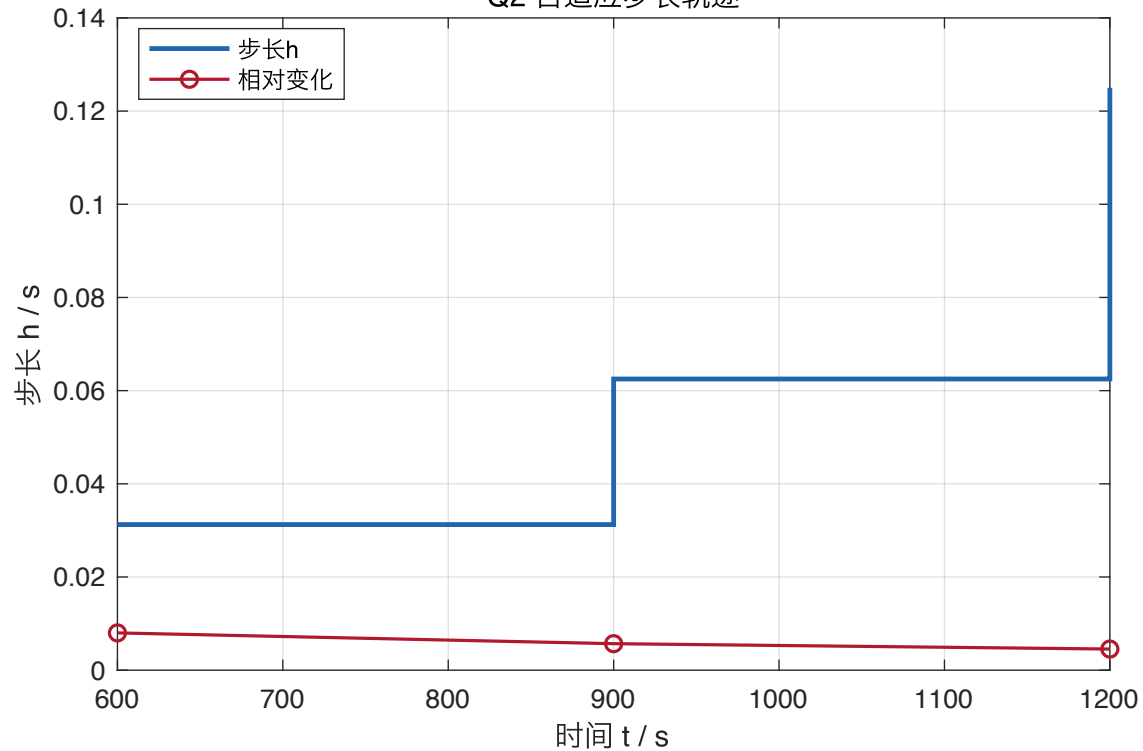


Q2 精度-效率帕累托图

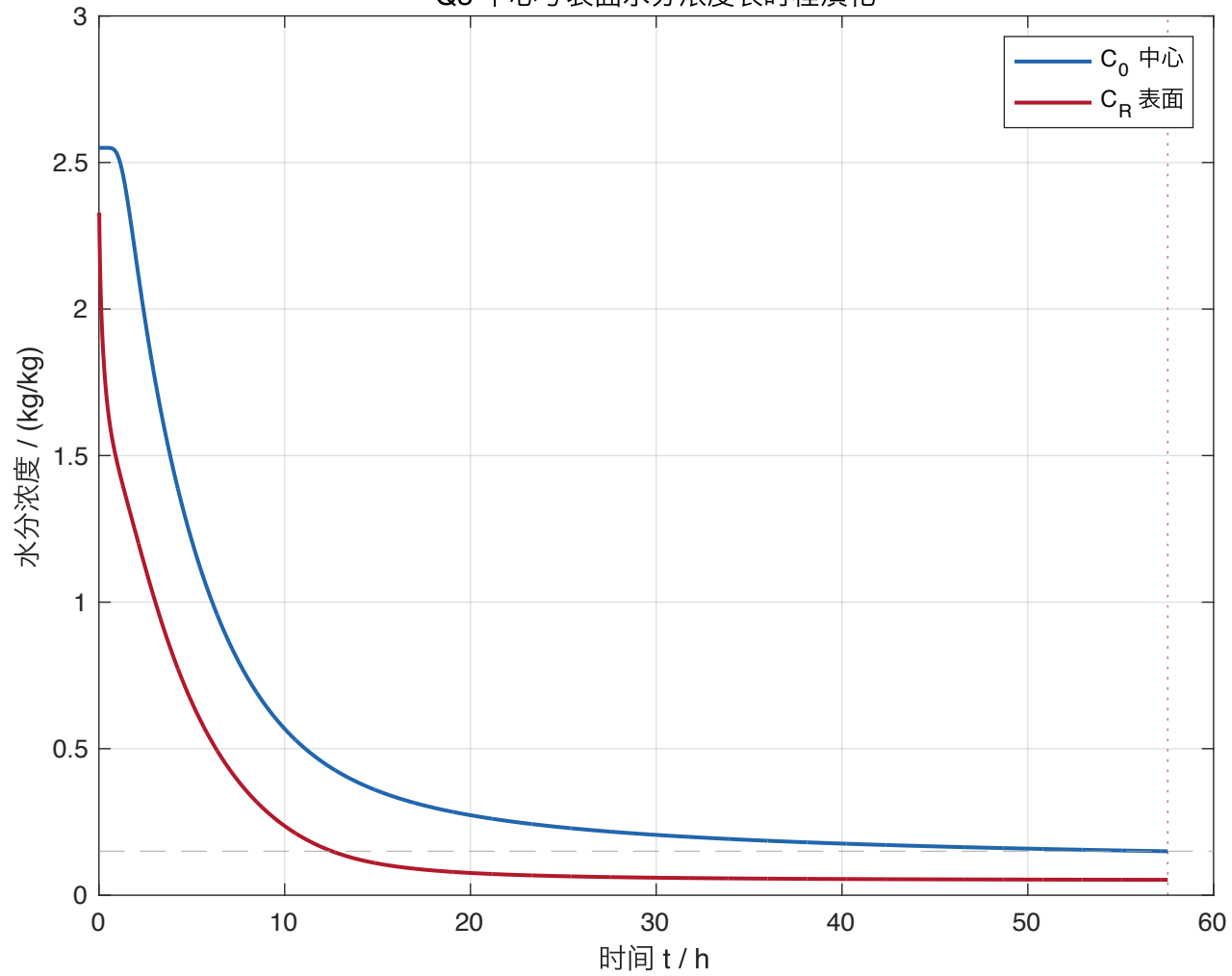


Q2 全局灵敏度 Sobol S_T 热力图

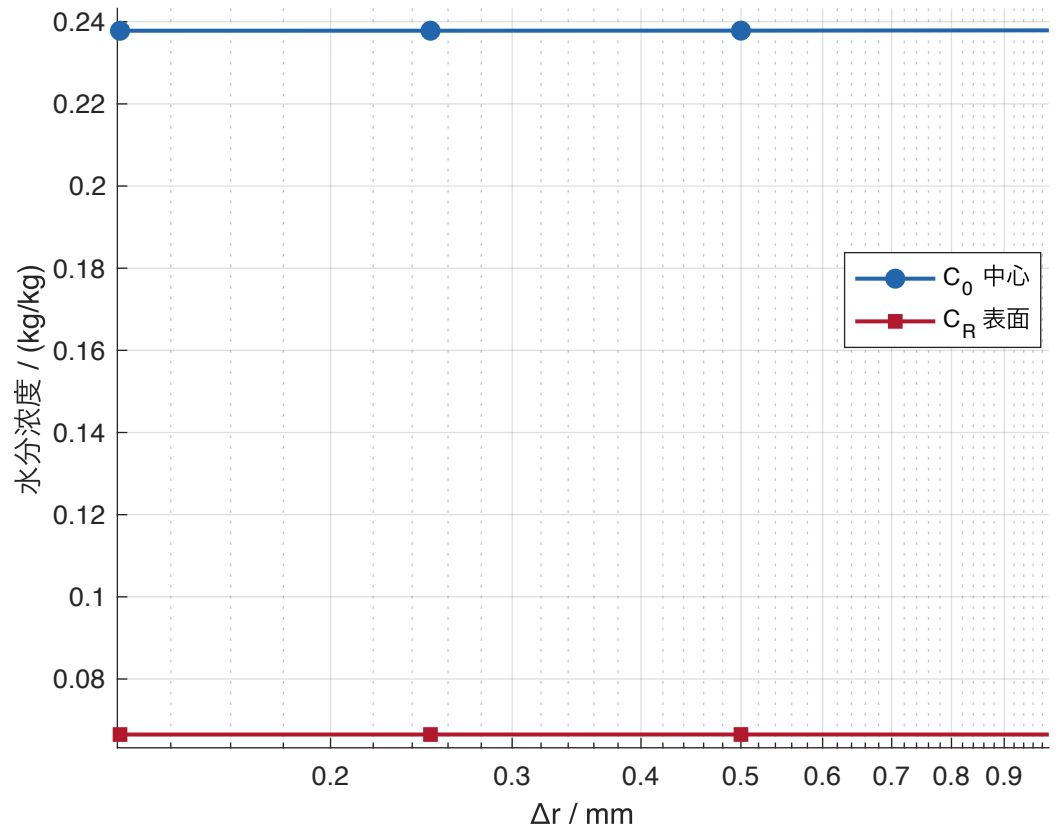
Q2 自适应步长轨迹



Q3 中心与表面水分浓度长时程演化

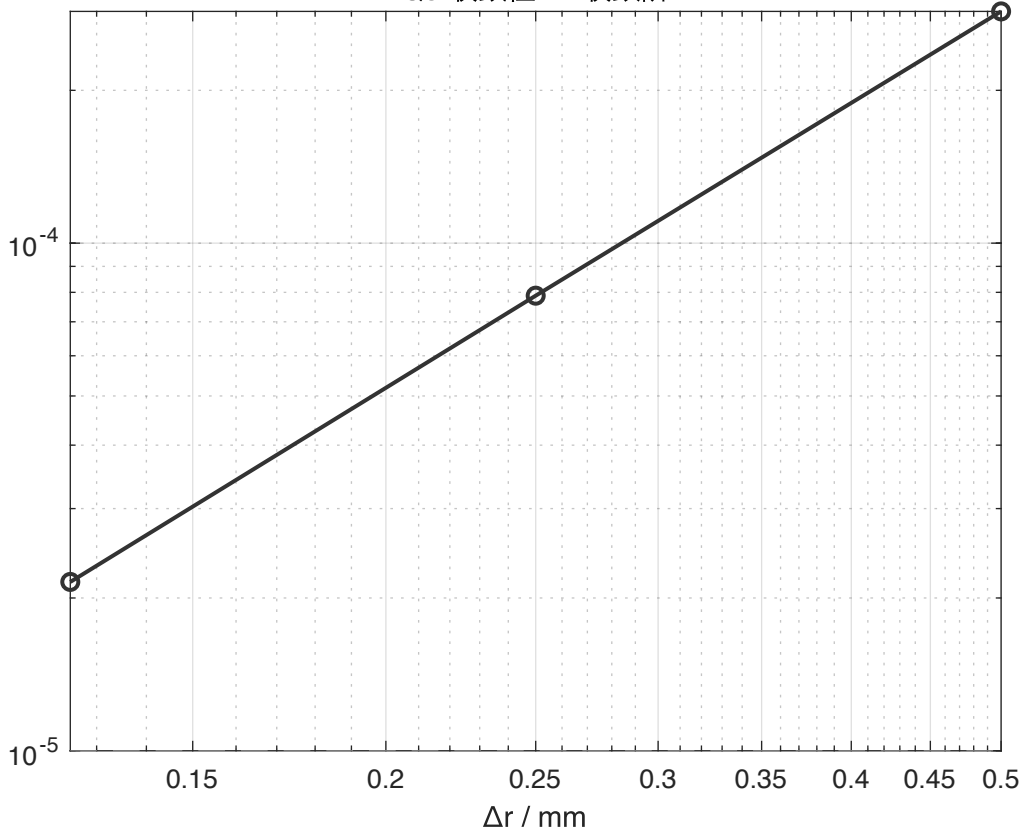


Q3 收敛性 — 终态值随步长

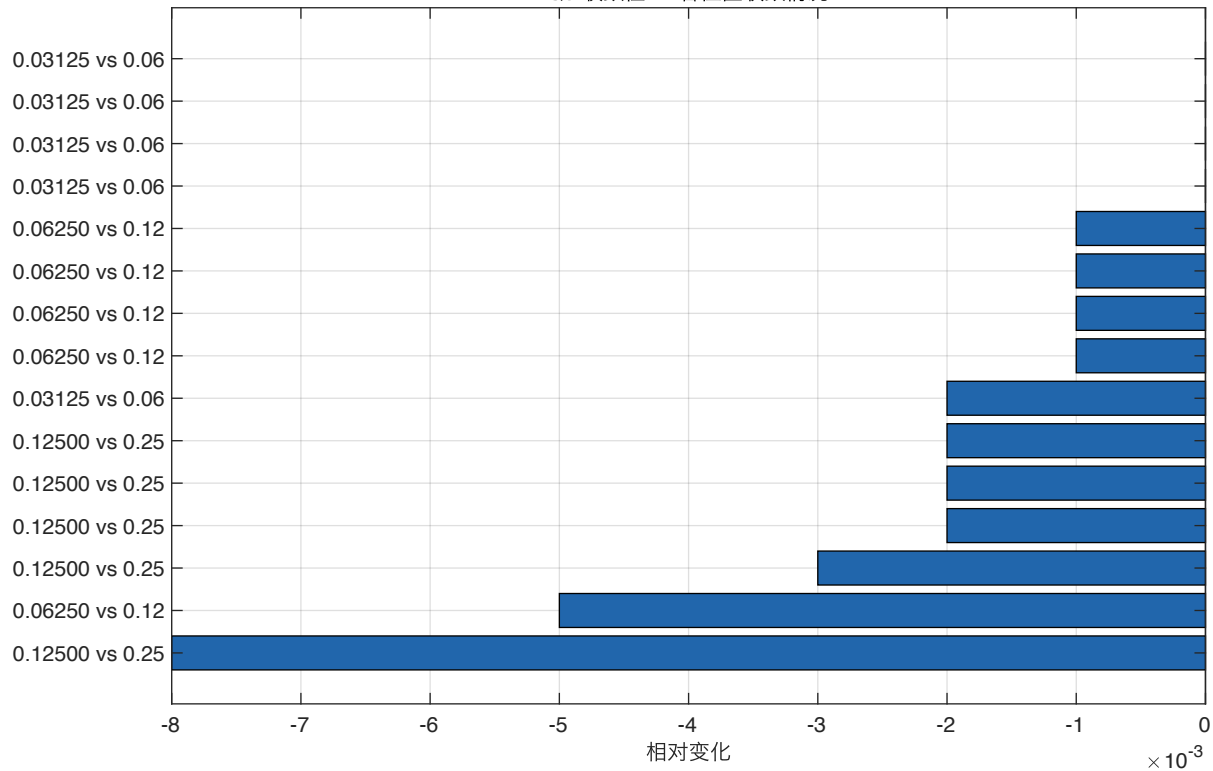


Q3 收敛性 — 收敛阶

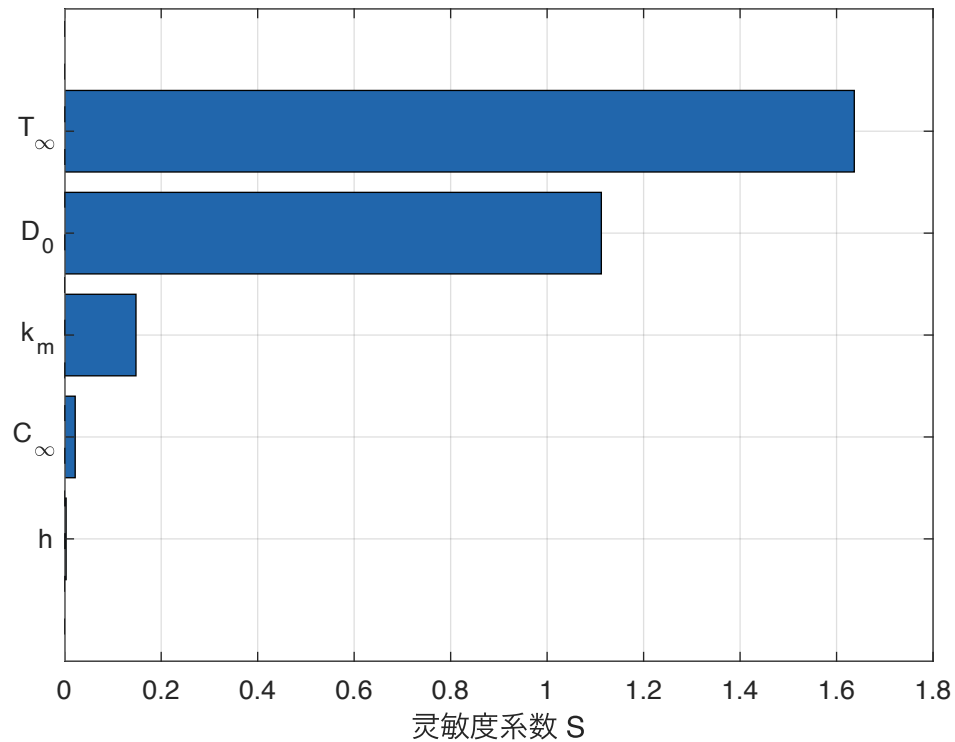
相对变化

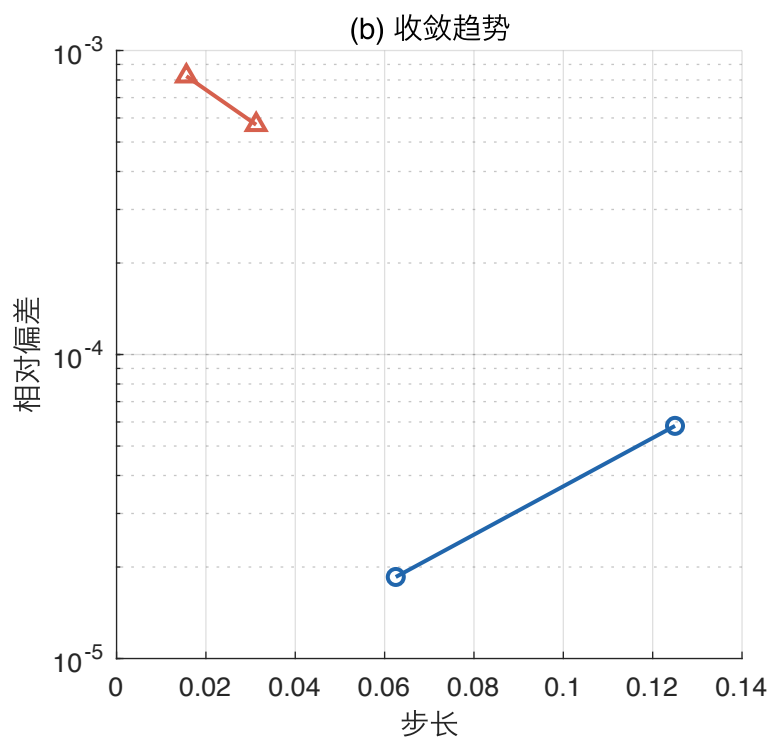
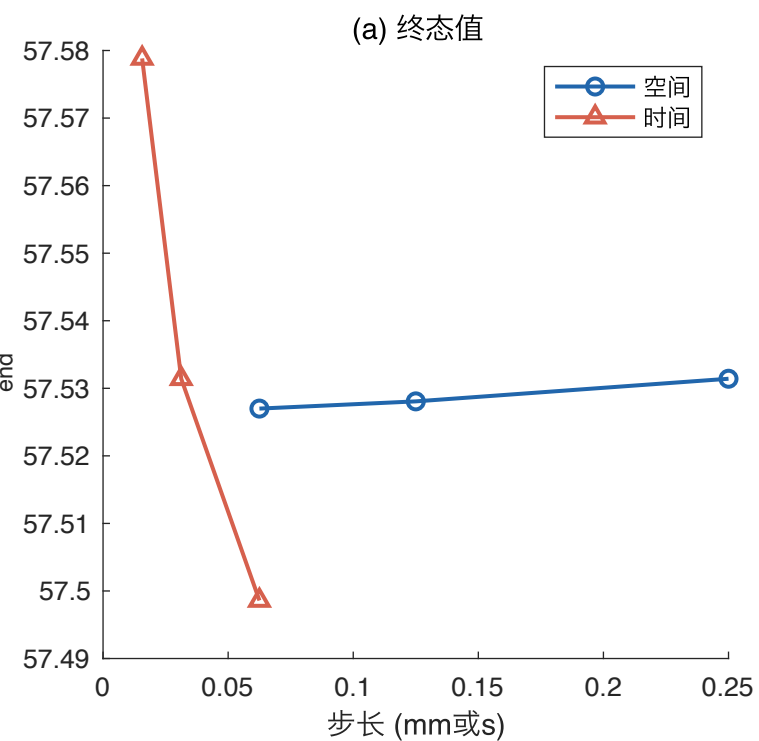


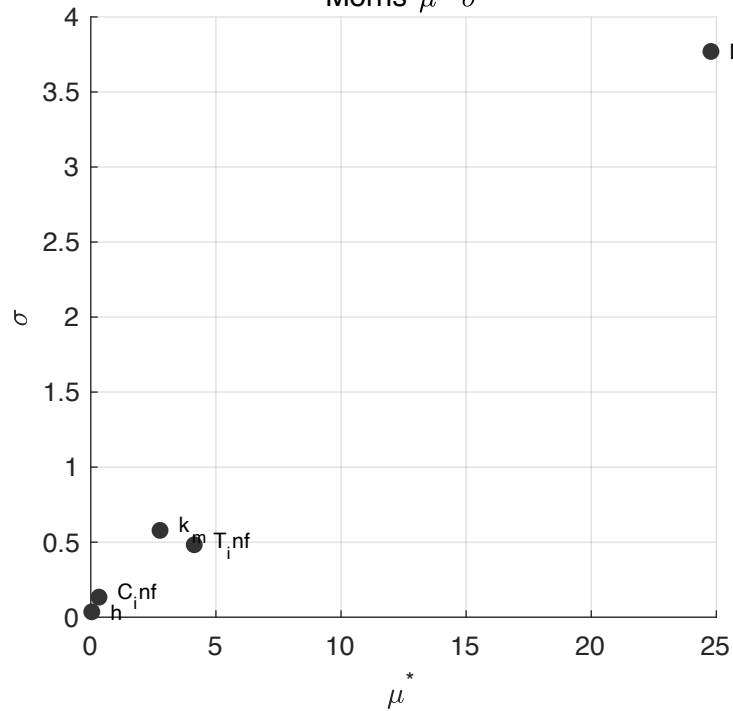
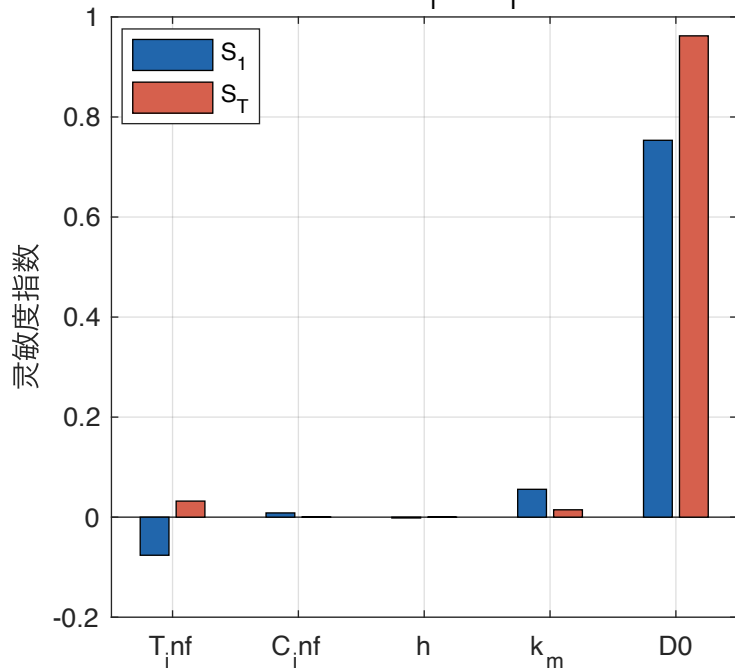
Q3 收敛性 — 各位置收敛情况



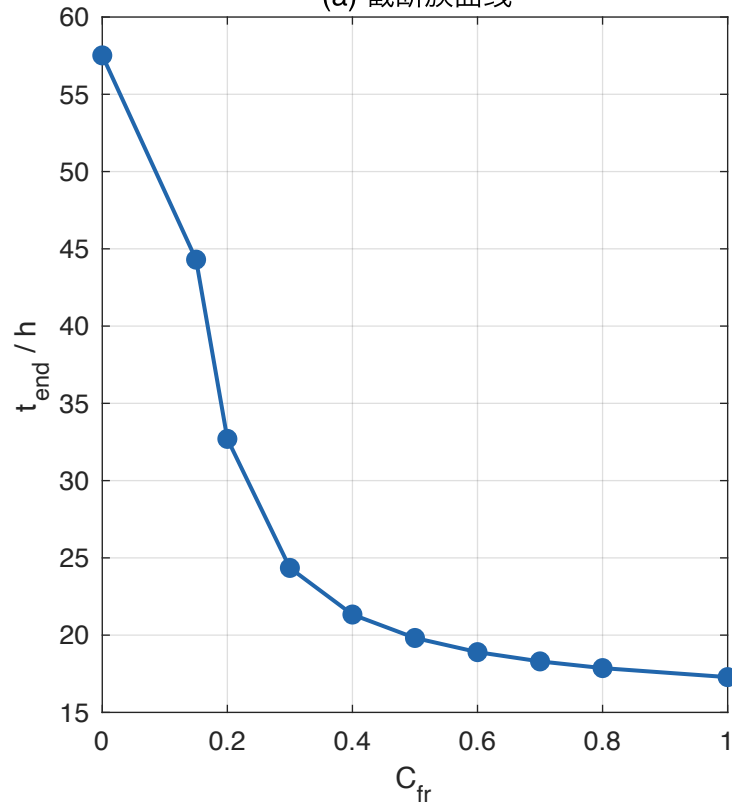
Q3 参数灵敏度



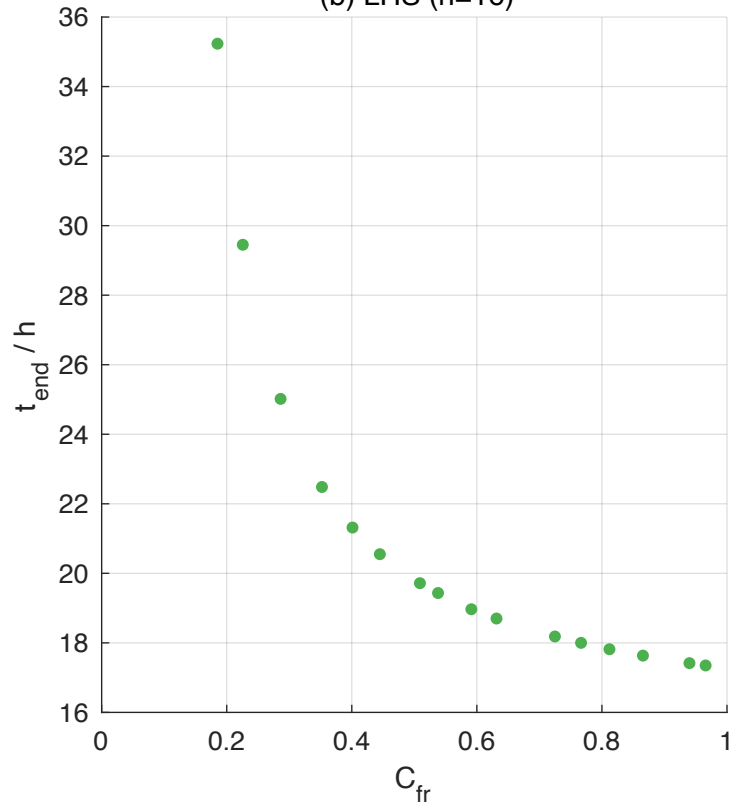


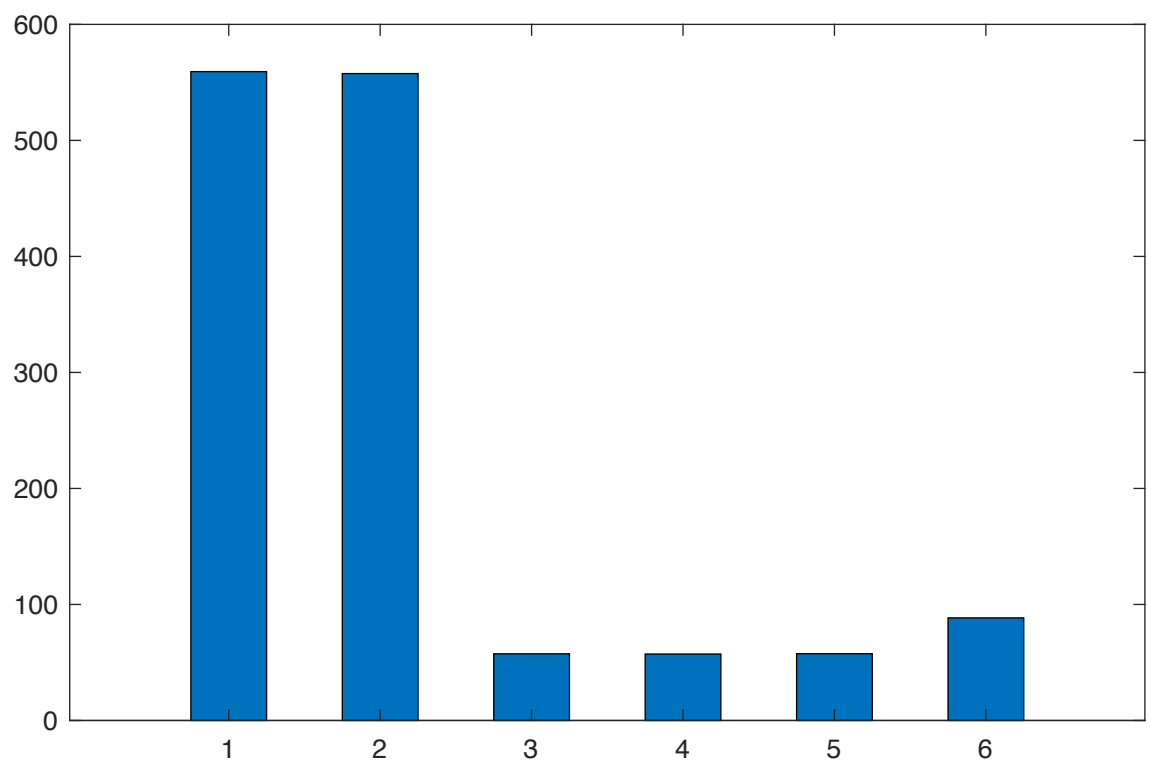
Morris $\mu^* - \sigma$ Sobol S_1 vs S_T 

(a) 截断族曲线

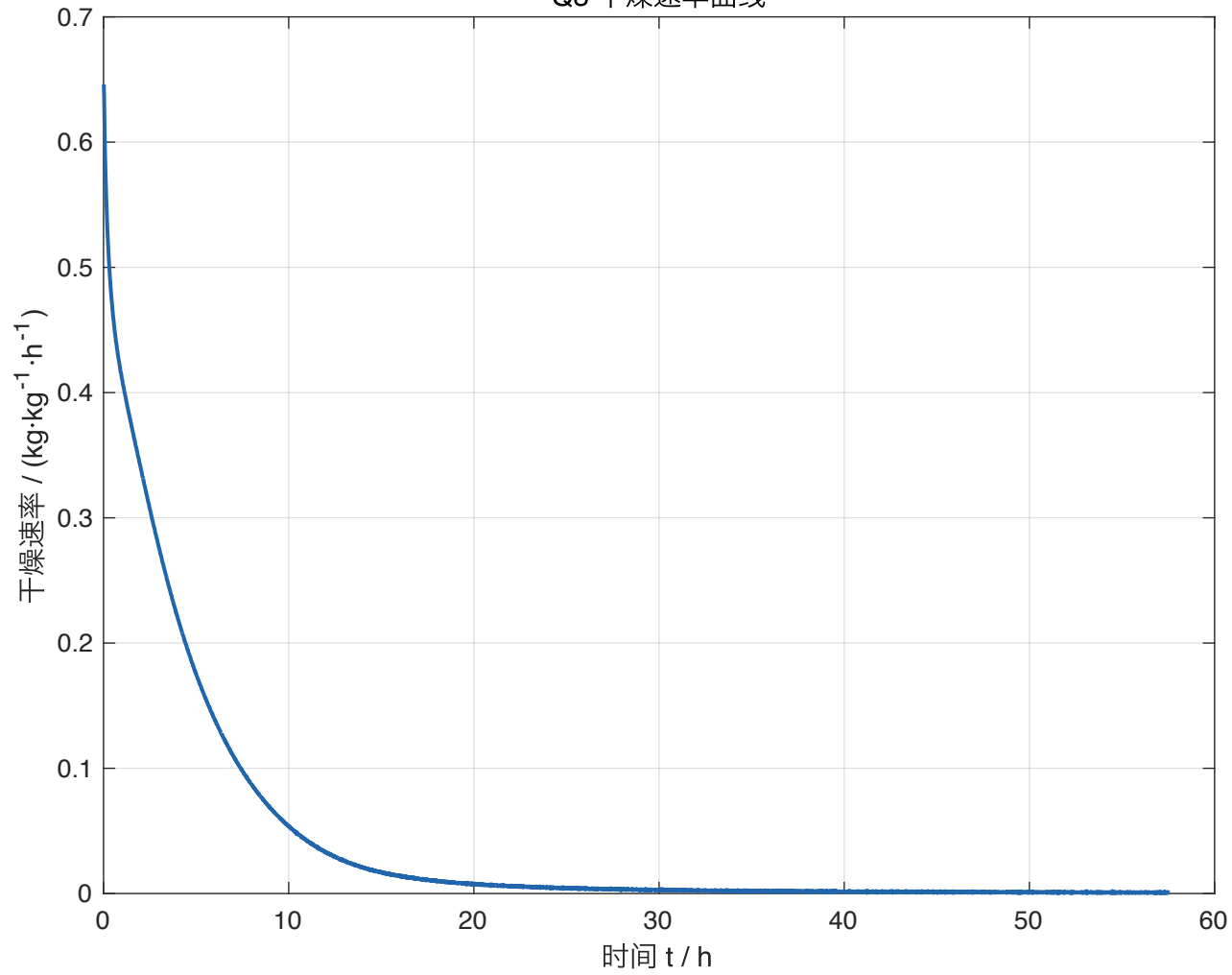


(b) LHS (n=16)



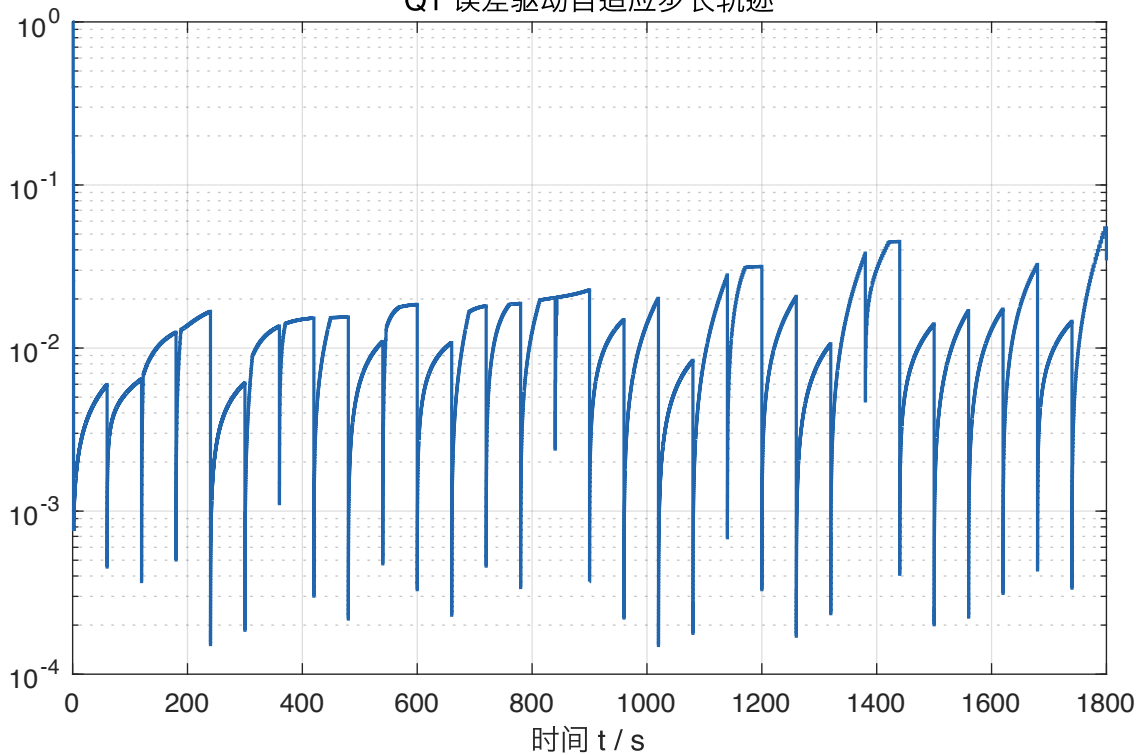


Q3 干燥速率曲线

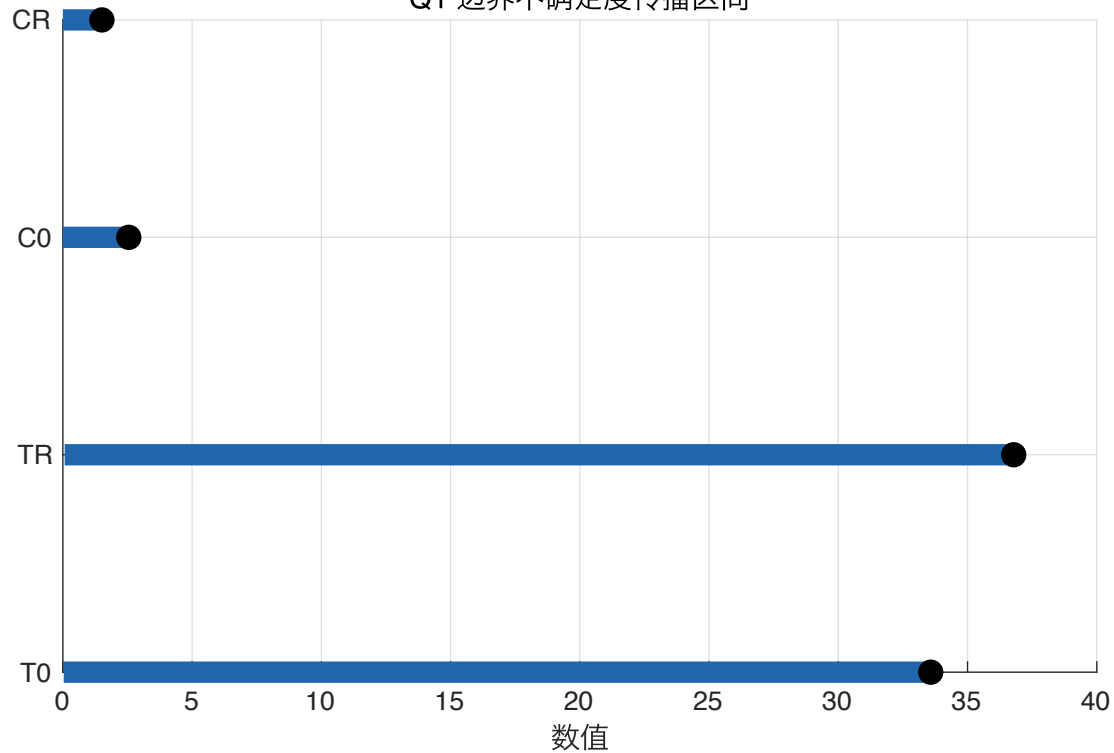


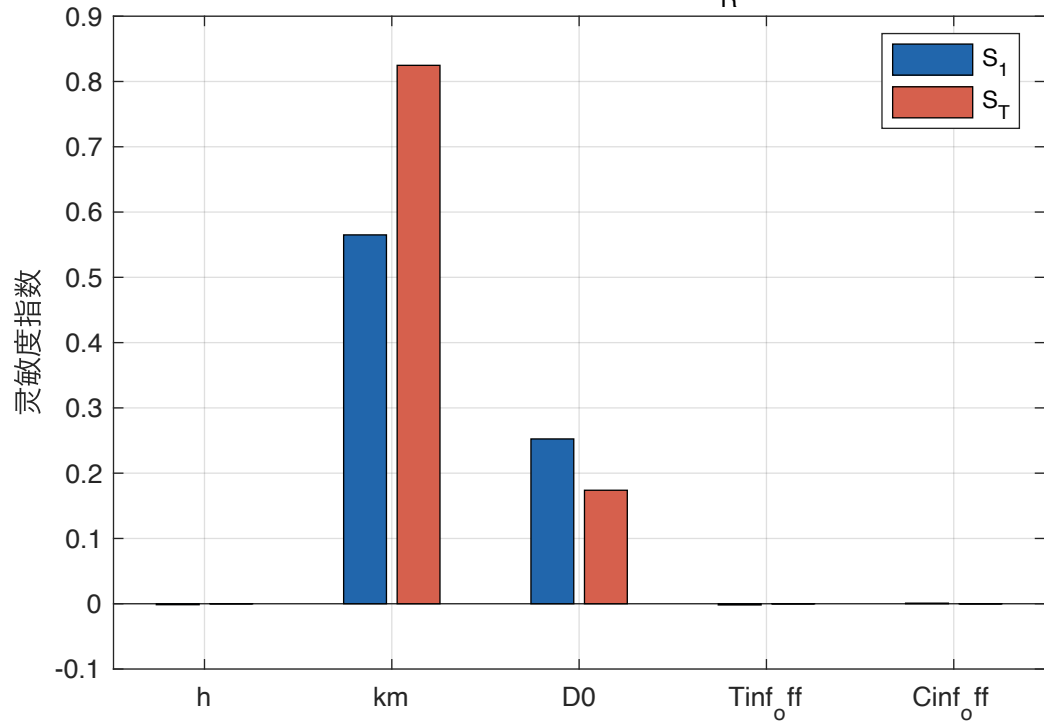
Q1 误差驱动自适应步长轨迹

步长 h/s

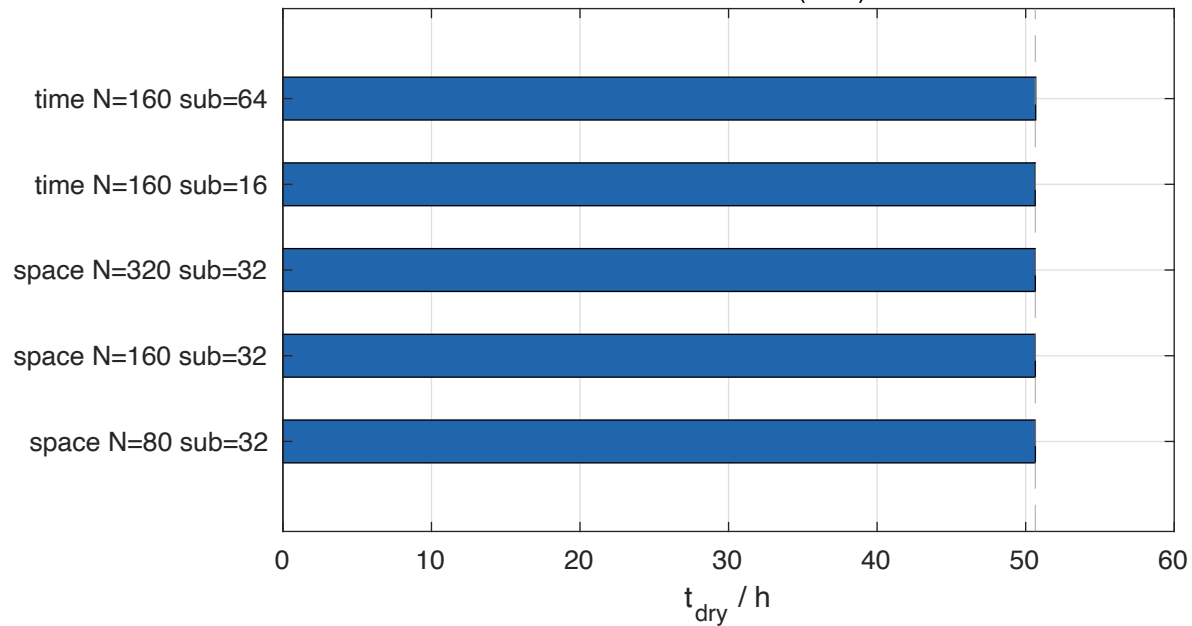


Q1 边界不确定性传播区间

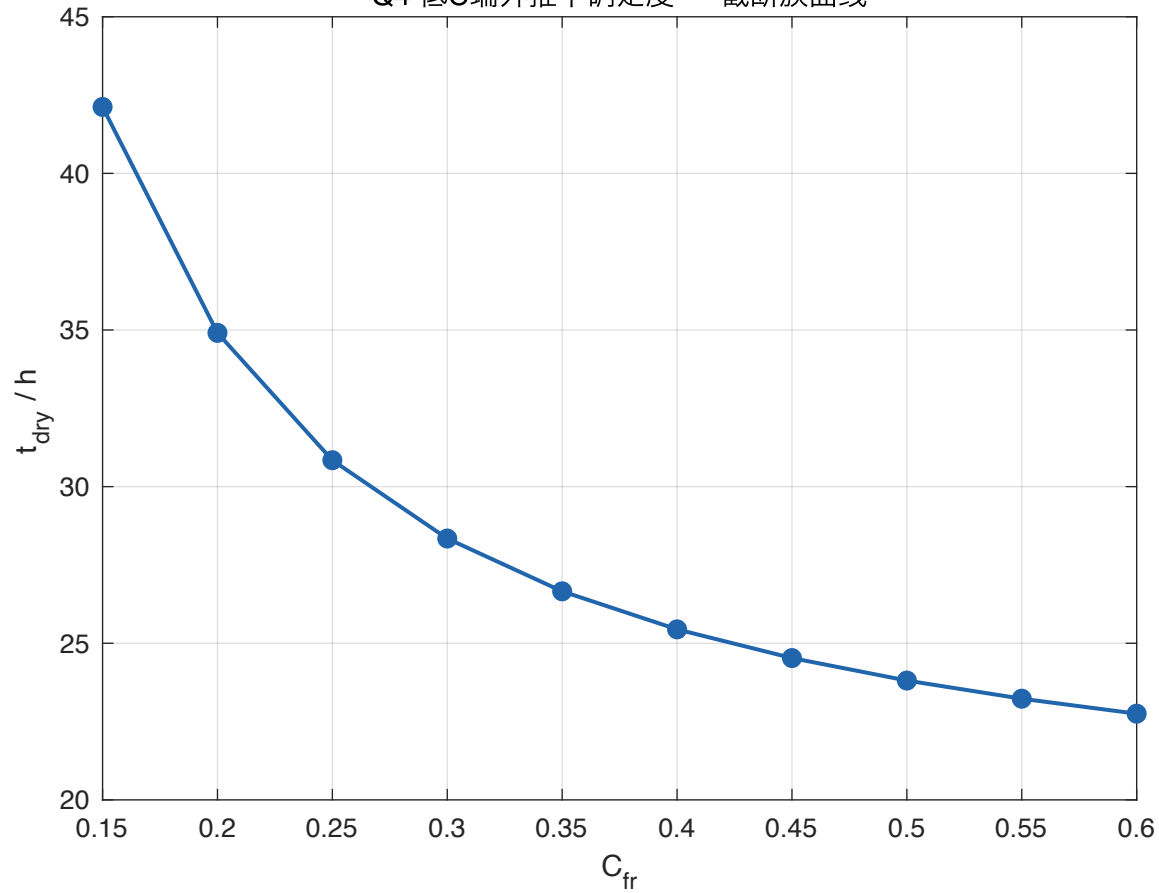


Q1 全局灵敏度 Sobol (C_R)

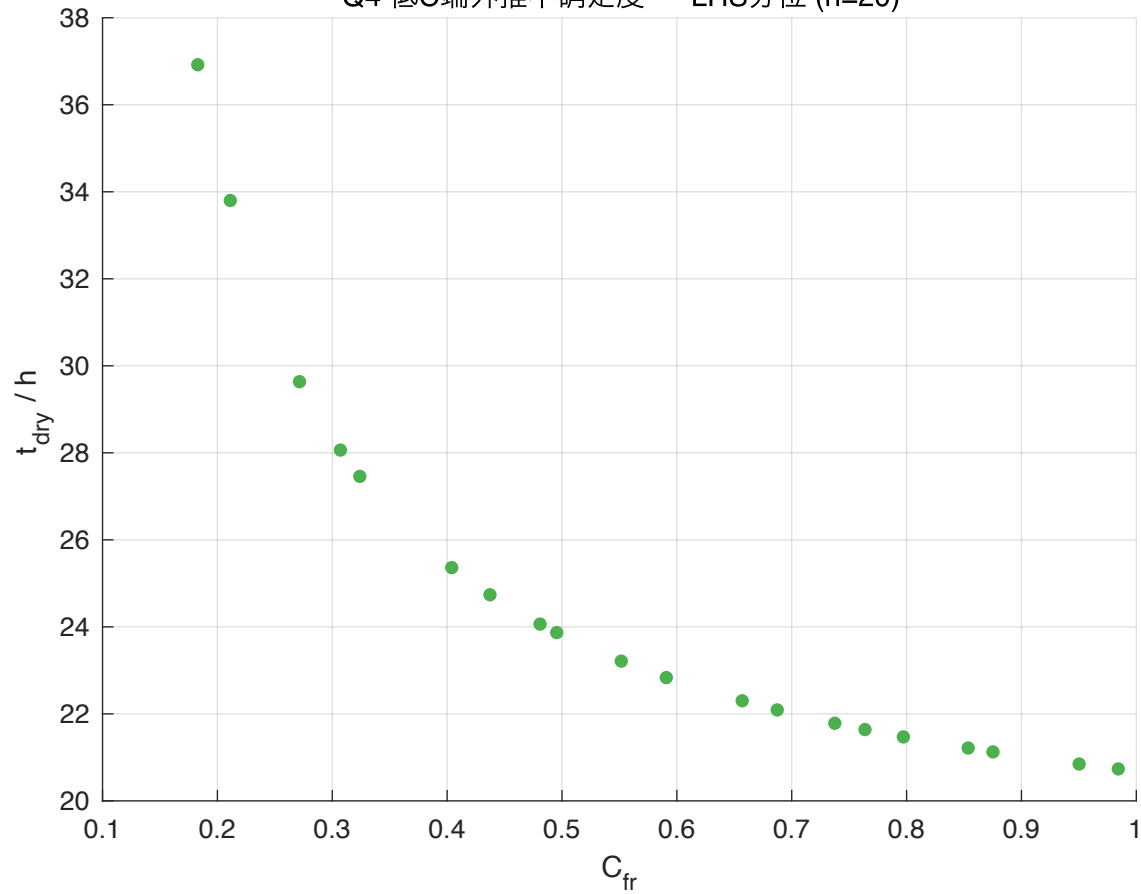
Q4 离散误差带 (GCI)



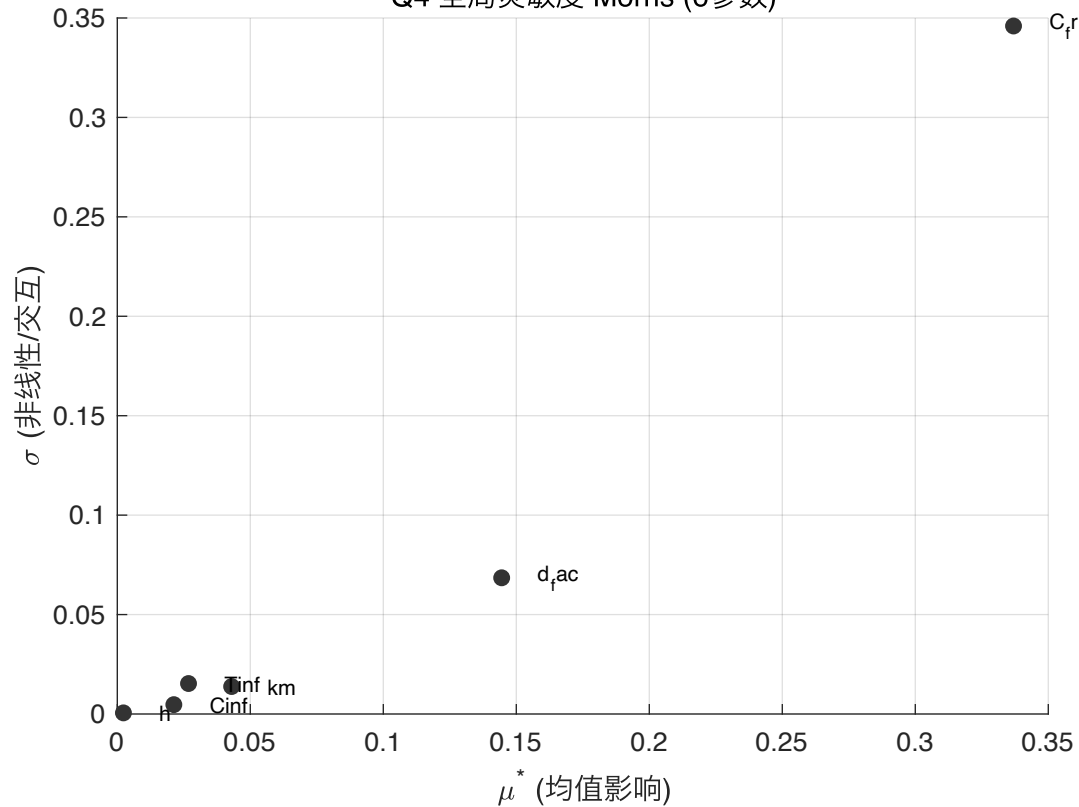
Q4 低C端外推不确定性 — 截断族曲线



Q4 低C端外推不确定度 — LHS分位 (n=20)

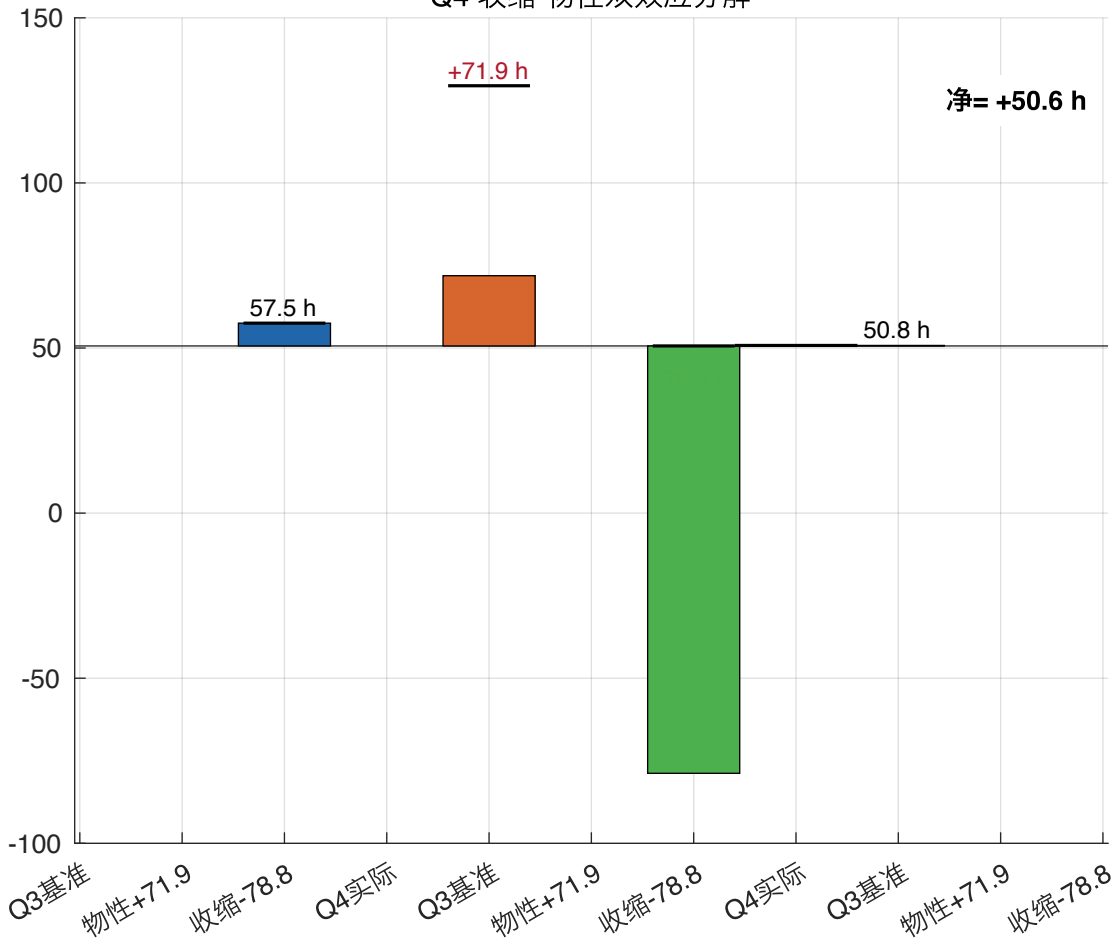


Q4 全局灵敏度 Morris (6参数)

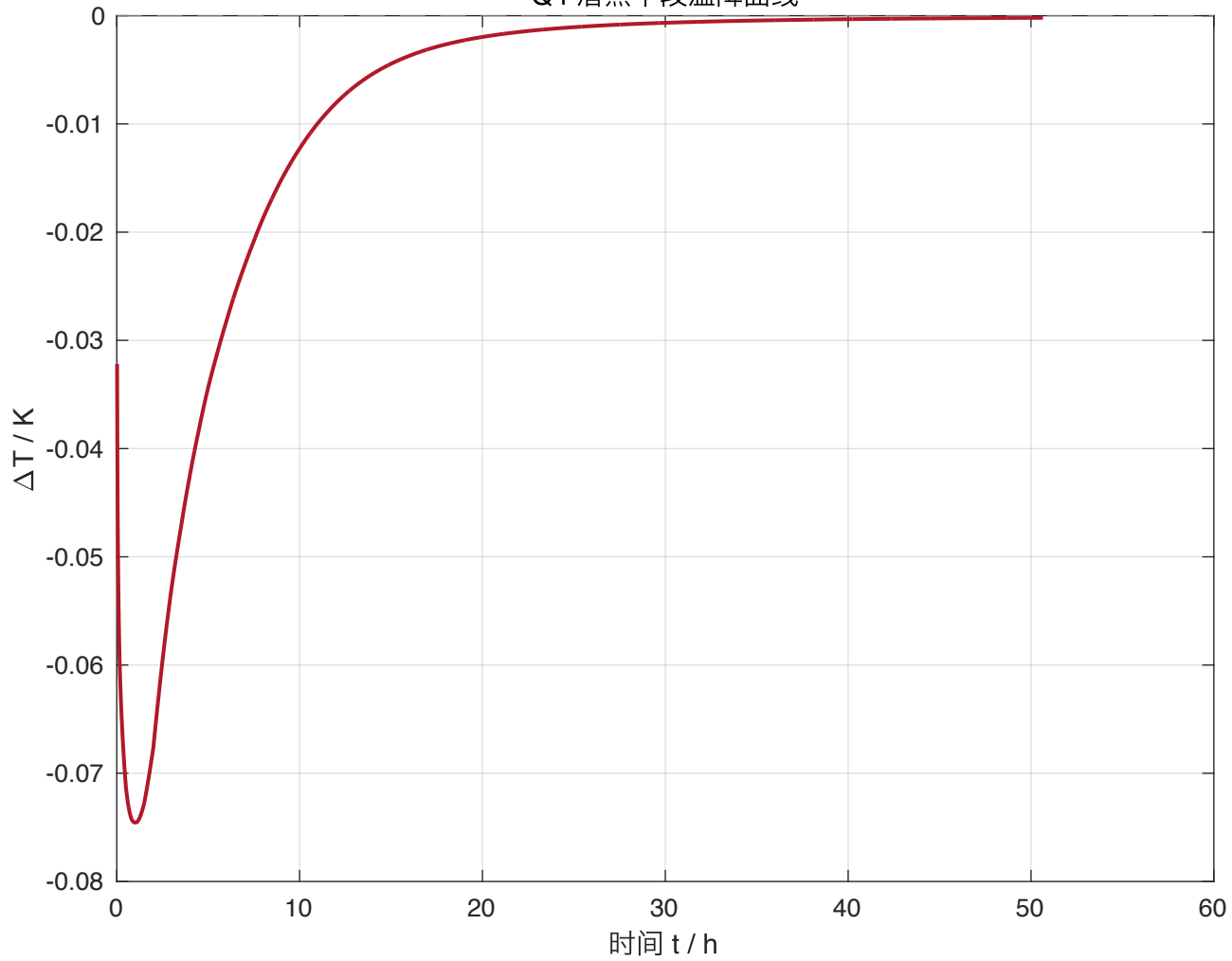


Q4 收缩-物性双效应分解

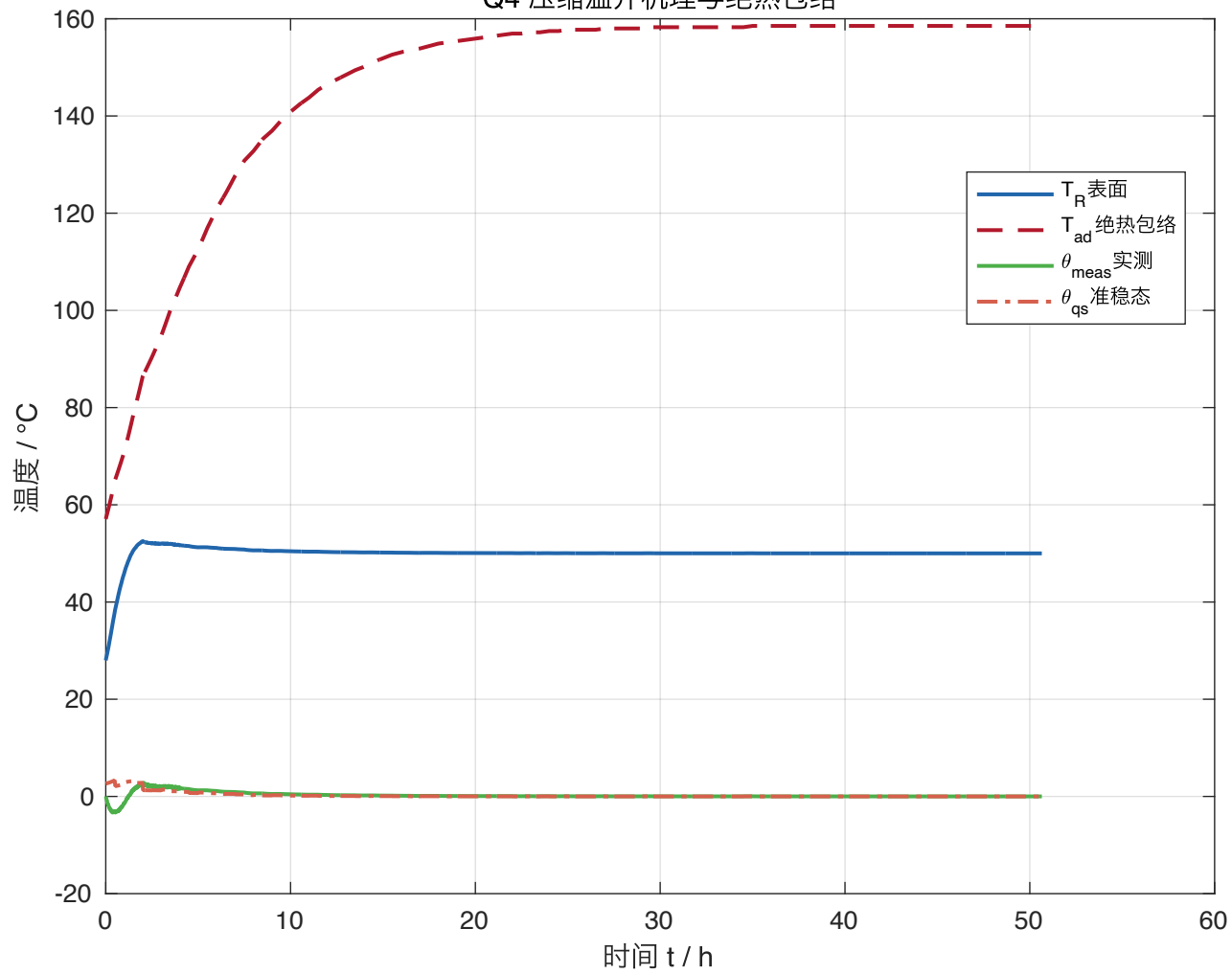
烘干时间 / h



Q4 潜热中段温降曲线



Q4 压缩温升机理与绝热包络



Q4 界面取法 x 网格对照

